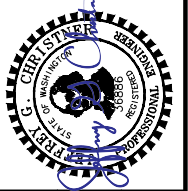


EASTSOUND SEWER AND WATER DISTRICT WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

NO.	REVISIONS	BY	DATE
1	NEW STAGING AREA	JGC	5/10/22
2	ADD INF. SAMPLE SUMP	JGC	5/10/22
3	ADD UV DISINFECTION	JGC	5/10/22
4	ADD TRENCH DRAINS	JGC	5/10/22
5	ADD INTERIM PLAN OF OP.	JGC	5/10/22
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



11-02-2022

AREA MAP- NOT TO SCALE



VICINITY MAP- NOT TO SCALE



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PROTECTION OF THE ENVIRONMENT

NO CONSTRUCTION-RELATED ACTIVITY SHALL CONTRIBUTE TO THE DEGRADATION OF THE ENVIRONMENT, ALLOW MATERIAL TO ENTER SURFACE OR GROUND WATERS, OR ALLOW PARTICULATE EMISSIONS TO THE ATMOSPHERE, WHICH EXCEED STATE OR FEDERAL STANDARDS. ANY ACTIONS THAT POTENTIALLY ALLOW A DISCHARGE TO STATE WATERS MUST HAVE PRIOR APPROVAL OF THE WASHINGTON STATE DEPARTMENT OF ECOLOGY.

PROJECT FUNDING INFORMATION

ECOLOGY SRF FUNDING NO. WQC-2021-EaSoWD-00208

CDBG FUNDING CONTRACT NO. 21-62210-036

CONTACT INFORMATION

CIVIL ENGINEER
WILSON ENGINEERING, LLC
JEFF CHRISTNER, PE & SAM RODRIGUEZ, EIT
PHONE 360.733.6100

OWNER
EASTSOUND SEWER AND WATER DISTRICT
JASON BRADSHAW, GENERAL MANAGER

PROJECT DATUM

HORIZONTAL: NAD 83/91

VERTICAL: NGVD 1929

THIS PROJECT IS FUNDED IN PART
BY THE WASHINGTON STATE
DEPARTMENT OF ECOLOGY



BID SET

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

COVER SHEET

DATE
NOV 2, 2022

SCALE
AS SHOWN

JOB NUMBER
2020-070

SHEET
G0.1

PAGE
1 OF 96

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NOTES & SPECIFICATIONS

GENERAL NOTES

1.

THE CONTRACTOR SHALL COORDINATE WITH EASTSOUND SEWER AND WATER DISTRICT ON ALL CHANGES TO EXISTING WASTEWATER FACILITY OPERATIONS. IT IS ESSENTIAL THAT EASTSOUND SEWER AND WATER DISTRICT OPERATIONS ARE NOT INTERRUPTED BY CONTRACTOR ACTIVITIES.
2.

CONTRACTOR AGREES THAT HE SHALL ASSUME SOLE AND COMPLETE RESPONSIBILITY FOR JOB SITE CONDITIONS DURING THE COURSE OF CONSTRUCTION OF THIS PROJECT, INCLUDING SAFETY OF ALL PERSONS AND PROPERTY, THAT THIS REQUIREMENT SHALL APPLY CONTINUOUSLY AND NOT BE LIMITED TO NORMAL WORKING HOURS, AND THAT THE CONTRACTOR SHALL DEFEND, INDEMNIFY, AND HOLD THE EASTSOUND SEWER AND WATER DISTRICT AND THE ENGINEER HARMLESS FROM ANY AND ALL LIABILITY, REAL OR ALLEGED, IN CONNECTION WITH THE PERFORMANCE OF WORK ON THIS PROJECT, EXCEPTING FOR LIABILITY ARISING FROM THE SOLE NEGLIGENCE OF THE EASTSOUND WATER AND SEWER DISTRICT OR THE ENGINEER.
3.

IF THERE ARE ANY DISCREPANCIES BETWEEN DIMENSIONS IN DRAWING AND EXISTING CONDITIONS WHICH WILL AFFECT THE WORK, THE CONTRACTOR SHALL BRING SUCH DISCREPANCIES TO THE ATTENTION OF THE ENGINEER FOR ADJUSTMENT BEFORE PROCEEDING WITH THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER FITTING OF ALL WORK AND FOR THE COORDINATION OF ALL TRADES, SUBCONTRACTORS, AND PERSONS ENGAGED UPON THIS CONTRACT.
4.

CONTRACTOR SHALL BE RESPONSIBLE TO REPAIR OR REPLACE ANY EXISTING IMPROVEMENTS OR UNDERGROUND FACILITIES DAMAGED BY HIM, HIS SUBCONTRACTORS, OR HIS MATERIAL SUPPLIERS WITHIN 48 HOURS OF THE DAMAGE OCCURRENCE AND/OR AS REQUIRED BY THE CONSTRUCTION INSPECTOR.
5.

EXISTING UTILITIES AND UNDERGROUND FACILITIES INDICATED ARE FOR INFORMATION ONLY. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY THE LOCATION AND DEPTH. NEITHER THE EASTSOUND SEWER AND WATER DISTRICT NOR THE ENGINEER ASSUMES RESPONSIBILITY THAT THE UTILITIES AND UNDERGROUND FACILITIES INDICATED WILL BE THE UTILITIES AND UNDERGROUND FACILITIES ENCOUNTERED.
6.

THE CONTRACTOR SHALL BE RESPONSIBLE FOR VERIFYING THE ELEVATIONS OF THE EXISTING STORM DRAINS, SEWERS, AND WATER TO BE EXTENDED, CROSSED, OR CONNECTED TO PRIOR TO COMMENCING THE WORK. NOTIFY ENGINEER IF ACTUAL IS DIFFERENT FROM PLANS.
7.

ALL TRENCHES AND EXCAVATIONS SHALL BE CONSTRUCTED IN STRICT COMPLIANCE WITH THE APPLICABLE SECTIONS OF FEDERAL OSHA REQUIREMENTS AND OTHER APPLICABLE SAFETY ORDINANCES. CONTRACTOR SHALL BEAR FULL RESPONSIBILITY FOR TRENCH SHORING DESIGN AND INSTALLATION.
8.

THE CONTRACTOR IS RESPONSIBLE FOR MATCHING EXISTING STREETS, SURROUNDING LANDSCAPE, AND OTHER IMPROVEMENTS WITH A SMOOTH TRANSITION IN PAVING, CURBS, GUTTERS, SIDEWALKS, GRADING, ETC, AND TO AVOID ANY ABRUPT OR APPARENT CHANGES IN GRADES OR CROSS SLOPES, LOW SPOTS, OR HAZARDOUS CONDITIONS.
9.

THE ENGINEER WILL PROVIDE CONSTRUCTION STACKING FOR THE PROJECT. SEE SUPPLEMENTAL CONDITIONS, NO. 7 SURVEYS, PERMITS, REGULATIONS.
10.

A 1-WEEK NOTIFICATION TO THE EASTSOUND SEWER AND WATER DISTRICT IS REQUIRED PRIOR TO ANY UTILITY SERVICE DISRUPTION. CONTRACTOR MUST PROVIDE TEMPORARY EQUIPMENT (POWER GENERATOR, PUMPS, PIPING, ETC) AND AN APPROVED PLAN FOR ALL WASTEWATER TREATMENT PLANT DISRUPTIONS. SEE SUMMARY OF WORK (SPECIFICATION SECTION 01 11 00.
11.

ALL APPROVALS AND PERMITS REQUIRED BY SAN JUAN COUNTY SHALL BE OBTAINED BY THE CONTRACTOR PRIOR TO THE START OF CONSTRUCTION.

SCHEDULE NOTES:

1.

ALL SITE EARTH WORK, CONCRETE WORK AND ASPHALT PAVEMENT REPAIR TO BE PRIORITIZED DURING DRY WEATHER MONTHS.

SURVEY MONUMENTATION:

CONTRACTOR SHALL BE FULLY RESPONSIBLE FOR OBTAINING PERMITS FROM THE WASHINGTON STATE DEPARTMENT OF NATURAL RESOURCES FOR REMOVING AND REPLACING ALL SURVEY MONUMENTATION THAT MAY BE AFFECTED BY CONSTRUCTION ACTIVITY, PURSUANT TO WAC 332-120. APPLICATIONS MUST BE COMPLETED BY A REGISTERED LAND SURVEYOR. APPLICATIONS FOR PERMITS TO REMOVE MONUMENTS MAY BE OBTAINED FROM THE WASHINGTON STATE DEPARTMENT OF NATURAL RESOURCES (DNR), OR BY CONTACTING THEIR OFFICE BY TELEPHONE AT (206) 902-1190.

WASHINGTON STATE DEPARTMENT OF NATURAL RESOURCES
PUBLIC LAND SURVEY OFFICE
1111 WASHINGTON STREET S.E.
OLYMPIA, WASHINGTON 98504-7060

UPON COMPLETION OF CONSTRUCTION, ALL MONUMENTS DISPLACED, REMOVED, OR DESTROYED SHALL BE REPLACED BY A REGISTERED LAND SURVEYOR, AT THE COST OF THE CONTRACTOR, PURSUANT TO THESE REGULATIONS. THE APPROPRIATE FORMS FOR REPLACEMENT OF SAID MONUMENTS SHALL BE COMPLETED AND FILED WITH DNR AT THE CONTRACTOR'S EXPENSE.

CONSTRUCTION INSPECTIONS:

CONTRACTOR MUST COORDINATE REQUIRED INSPECTIONS WITH THE COUNTY BUILDING INSPECTION OFFICIALS. IN ADDITION, CONTRACTOR IS TO PROVIDE OWNER AND ENGINEER 24-HOUR (MINIMUM) NOTICE FOR SCHEDULED INSPECTIONS. REQUIRED INSPECTIONS INCLUDE THE FOLLOWING ITEMS:

1.

BUILDING FOOTINGS
2.

CONCRETE STRUCTURES
3.

ELECTRICAL INSPECTIONS
4.

FINAL OVERALL INSPECTION

CONTRACTOR MUST PROVIDE OWNER AND ENGINEER 24-HOUR (MINIMUM) NOTICE FOR ALL SPECIAL INSPECTIONS IDENTIFIED ON S1.1 - SPECIAL INSPECTIONS, TESTING, AND STRUCTURAL OBSERVATIONS.

GENERAL NOTES, CONT.

PROTECTION OF THE ENVIRONMENT

NO CONSTRUCTION RELATED ACTIVITY SHALL CONTRIBUTE TO THE DEGRADATION OF THE ENVIRONMENT, ALLOW MATERIAL TO ENTER SURFACE OR GROUND WATERS, OR ALLOW PARTICULATE EMISSIONS TO THE ATMOSPHERE, WHICH EXCEED STATE OR FEDERAL STANDARDS. ANY ACTIONS THAT POTENTIALLY ALLOW A DISCHARGE TO STATE WATERS MUST HAVE PRIOR APPROVAL OF THE STATE OF WASHINGTON, DEPARTMENT OF ECOLOGY

OPERATION PLAN

WWTP UGRADE CONSTRUCTION - GENERAL NOTES

- 1)

THE EXISTING WASTEWATER TREATMENT PLANT (WWTP) SHALL NOT BE INTERRUPTED DURING CONSTRUCTION EXCEPT AS DESCRIBED BELOW. THE CONTRACTOR SHALL BE RESPONSIBLE FOR SEQUENCING CONSTRUCTION ACTIVITIES SO THAT NO INTERRUPTION OCCURS.
- 2)

THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING ALL TEMPORARY BY-PASS PUMPING, PIPING, HOLDING TANKS AND ADDITIONAL STORAGE NECESSARY TO KEEP SEWER AND WWTP SYSTEM UNINTERRUPTED DURING CONSTRUCTION ACTIVITIES.
- 3)

THE PLAN ASSUMES THAT ALL BASIN, BUILDINGS, EQUIPMENT, AND PIPING WILL BE INSTALLED /TESTED AND READY TO BE CONNECTED PRIOR TO SWITCH-OVER. THIS IS A SUGGESTED PLAN AND DOES NOT RELIEVE THE CONTRACTOR FROM OBLIGATIONS TO AVOID INTERRUPTION:
- 4)

EMERGENCY CONTACTS:

ESWD: JASON BRADSHAW, GENERAL MANAGER, CELL: 620-441-4006
WILSON ENG: JEFF CHRISTNER, PROJECT ENGINEER, CELL: 360-303-5129

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



WILSONENGINEERING.COM



DESIGNED BY	JGC	DRAWN BY	SNR	CHECKED BY	AWL
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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

NOTES & SPECIFICATIONS

SHEET	DATE	SCALE	JOB NUMBER
G0.3	NOV 2, 2022	AS SHOWN	2020-070
PAGE			
3		OF	96



BID SET

11-02-2022

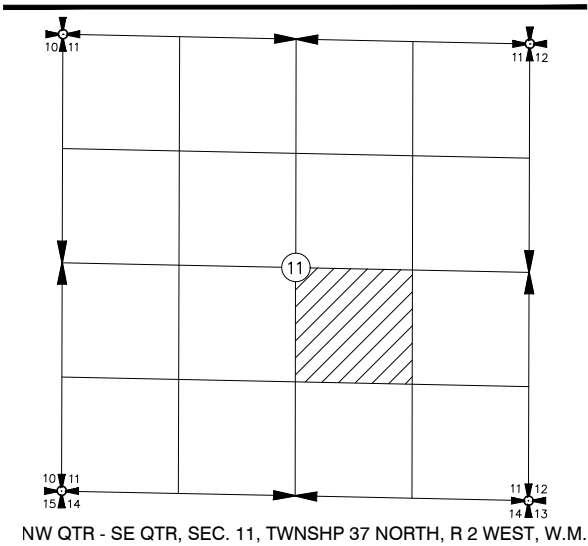
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EASTSOUND SEWER AND WATER DISTRICT

WASTEWATER TREATMENT PLANT DESIGN

W.A.C. 332-130 COMPLIANCE SHEET

SECTIONAL INDEX DATA



NOTICE TO USER

EFFECTIVE JANUARY 13, 2019, ALL TOPOGRAPHIC MAPS PREPARED BY A LICENSED SURVEYOR IN THE STATE OF WASHINGTON, AND SUBJECT TO THE LICENSURE AND PRACTICE REQUIREMENTS ESTABLISHED BY THE WASHINGTON STATE BOARD OF REGISTRATION FOR ENGINEERS AND LAND SURVEYORS, MUST INCLUDE THE DESCRIPTIVE NOTES AND METADATA ENUMERATED UNDER W.A.C 332-130-145 AND ITS APPURTENANT SECTIONS OF 332-130. THIS EXHIBIT IS INTENDED TO ADDRESS THE STATUTORY REQUIREMENTS STIPULATED BY THIS W.A.C DIRECTIVE.

W.A.C. 332-130-145 REQUIRED DATA

1.E: THIS SURVEY WAS PREPARED UNDER THE DIRECT SUPERVISION OF:

PAUL J. DARROW, WA PLS #50697
SR. PROJECT SURVEYOR
WILSON ENGINEERING LLC
805 DUPONT STREET, SUITE 7
BELLINGHAM, WA 98225
360-733-6100 (EXT. 243)
pdarrow@wilsonengineering.com

2.A: BASIS OF ELEVATIONS: ELEVATION VALUES AND CONTOURS DEPICTED ON THIS SURVEY ARE BASED UPON HOLDING AS FIXED THE NGVD29 DATUM, PER BENCHMARK NO. 20 RLM, ACCORDING TO THE PERPETUATION OF SAN JUAN COUNTY BENCHMARK #00165, HAVING AN ELEVATION OF 34.28, PER LAND CORNER RECORD # 2000 0303018.

2.B: PURPOSE OF SURVEY: WILSON ENGINEERING PERFORMED THIS SURVEY DURING JANUARY OF 2013 WITH ADDITIONAL DATA AND CHANGES BEING SURVEYED IN MARCH OF 2021, AT THE REQUEST OF EASTSOUND SEWER AND WATER DISTRICT PURSUANT TO WASTEWATER TREATMENT PLANT UPGRADE DESIGN. THIS SURVEY WAS PREPARED WITH THE BENEFIT OF A TITLE REPORT, AND THE DEPICTED PARCEL BOUNDARY SHOULD BE CONSIDERED AUTHORITATIVE.

2.C: SOURCE OF CONTOURS: THE CONTOURS DEPICTED ON THIS SURVEY WERE DERIVED BASED ON DIRECT FIELD OBSERVATIONS.

2.D: CONTOUR INTERVAL LABELING: MAJOR CONTOURS AT 1-FOOT INTERVALS HAVE BEEN EXPLICITLY LABELED.

2.E: DESCRIPTION OF BENCHMARKS SET PURSUANT TO THIS SURVEY: REFER TO THE ACCOMPANYING "CONTROL TABLE" FOR COORDINATES, ELEVATION, AND DESCRIPTION OF ON-SITE CONTROL SET PURSUANT TO THIS SURVEY.

2.F: ELEVATION AND/OR CONTOUR ACCURACY: IF CONTOURS HAVE BEEN DEPICTED ON THE FACE OF THIS SURVEY, IT IS ANTICIPATED THAT 90% OF ANY MEASURED ELEVATION VALUE, IF OBSERVED RELATIVE TO THE CONTROL POINTS SPECIFICALLY ENUMERATED IN THE ACCOMPANYING CONTROL TABLE, WILL BE, IN FACT, WITHIN ONE-HALF OF THE MINOR-CONTOUR INTERVAL DEPICTED HEREON. SPECIFIC ELEVATIONS DEPICTED HEREON, IF ANY, ARE EXPECTED TO BE WITHIN ONE INTEGRAL VALUE OF THE FINAL DEPICTED SIGNIFICANT FIGURE. THAT IS, 90% OF ELEVATIONS EXPRESSED TO THE TENTH-FOOT, SHOULD BE WITHIN 0.1 FEET OF THAT VALUE, IF OBSERVED RELATIVE TO THE SURVEY CONTROL SPECIFICALLY ENUMERATED IN THE ACCOMPANYING CONTROL TABLE. IF OFF-SITE CONTROL IS EMPLOYED, EVEN CONTROL PURPORTING TO BE ON THE SAME DATUM OR BASED ON THE SAME OFF-SITE BENCHMARK, THEN NO ABSOLUTE STATEMENT REGARDING THE ACCURACY OF THE DEPICTED POINTS CAN BE MADE, AND VALUES SO OBSERVED ARE OUTSIDE OF THIS SURVEY'S AUTHORITY OR INTEREST.

2.G: STATEMENT OF USE: AS NOTED IN SECTION 2.B, THIS SURVEY WAS PREPARED FOR THE SPECIFIC PURPOSE OF WASTEWATER TREATMENT PLAN UPGRADE DESIGN. IN THE COURSE OF PREPARING THIS SURVEY, PURSUANT TO THIS PURPOSE, ANCILLARY DATA NECESSARY TO ACCOMPLISH THIS SURVEYS INTENDED PURPOSE MAY HAVE BEEN CAPTURED. IN THE CASE OF THIS SURVEY A NUMBER OF BUILDING ENVELOPES ON ADJACENT PARCELS WERE CAPTURED, BUT THE DEPICTION OF SAME SHOULD NOT BE CONSIDERED AUTHORITATIVE.

2.H: SOURCE OF CONTROLLING BOUNDARY INFORMATION: THE OWNERSHIP BOUNDARIES DEPICTED ON THIS SURVEY ARE BASED UPON SOME, OR ALL, OF THE DOCUMENTS ENUMERATED IN THE ACCOMPANYING "REFERENCE DOCUMENTS" AS THEREIN CHARACTERIZED. BEARINGS HAVE BEEN TRANSLATED AND/OR ROTATED FROM THE RECORD VALUES TO CONFORM WITH THE CURRENT DATUM OF THIS SURVEY.

3.A: SOURCE OF DEPICTED UTILITY INFORMATION: UTILITY LINES DEPICTED ON THIS SURVEY ARE BASED UPON DIRECT OBSERVATION OF SURFACE VISIBLE STRUCTURES AND LOCATE MARKS MADE BY UTILITY LOCATING PROFESSIONALS.

3.B: ACCURACY OF DEPICTED UTILITY INFORMATION: WILSON ENGINEERING DOES NOT PROVIDE FOR-HIRE UTILITY LOCATION AND/OR MARKING SERVICES, AND CAN NOT INDEPENDENTLY ASCERTAIN THE ACCURACY OF ANY DEPICTED UTILITY THAT WAS NOT DIRECTLY OBSERVED IN THE COURSE OF THIS SURVEY.

3.C: STATEMENT OF LIMITATIONS REGARDING UTILITY-DEPICTION ACCURACY: EASTSOUND SEWER AND WATER DISTRICT HAS BEEN NOTIFIED THAT WILSON CAN NOT, AND DOES NOT, GUARANTEE THE ACCURACY, AT ANY LEVEL, OF DEPICTED UTILITIES BASED ON THIRD-PARTY PAINT MARKS OR RECORD INFORMATION.

CONTROL NOTES

HORIZONTAL DATUM: BASIS OF COORDINATES, BEARINGS AND CALCULATIONS ARE NAD 83/91 WASHINGTON STATE PLANE NORTH ZONE BASED ON TIES TO MONUMENTS AS SHOWN PER RECORD OF SURVEY ON FILE IN BOOK 16 OF SURVEYS AT PAGE 99, RECORDS OF SAN JUAN COUNTY, WASHINGTON.

BASIS OF COORDINATES: COORDINATION AND MENSURATION ARE LOCAL GROUND VALUES, BASED UPON HOLDING THE PUBLISHED NAD83/91 POSITION FOR THE 3" BRASS DISC IN CASE AT PROJECTED CENTERLINE OF MT. BAKER ROAD, SOUTH OF THE AIRPORT CENTER BUILDING. DISC BEARS THE STAMP "SAN JUAN COUNTY SURVEY CONTROL MONUMENT 00165". SAID MONUMENT HAS THE FOLLOWING PUBLISHED POSITION:

NORTHING = 628,007.20 USFT
EASTING = 1,138,590.22 USFT

BASIS OF BEARINGS: BEARINGS ARE BASED UPON HOLDING THE CALCULATED INVERSE BETWEEN SAN JUAN COUNTY MONUMENTS "00165" AND THE BRASS DISK PRESUMED TO BE THAT MONUMENT MARKING THE E 16TH CORNER BETWEEN SECTIONS 11 AND 14, PER THE AFOREMENTIONED RECORD OF SURVEY. SAID INVERSE BEING S 88° 16' 54" E, AT A DISTANCE OF 1301.67' (GRID). THE PUBLISHED WASHINGTON STATE PLANE (NORTH ZONE) VALUES FOR THE 16TH-CORNER MONUMENT ARE AS FOLLOWS:

NORTHING = 627,968.17 USFT
EASTING = 1,139,891.30 USFT

VERTICAL DATUM:
NGVD 29

BASIS OF ELEVATIONS: ELEVATIONS SHOWN ON THIS SURVEY ARE BASED ON THE USGS BENCH MARK DESIGNATED 20 RLM (BRASS DISC IN CONCRETE) AT THE SOUTH END OF THE RUNWAY AT ORCAS ISLAND AIRPORT, HAVING AN ELEVATION OF 31.0 FEET (NGVD 29), PER THE PERPETUATION OF SAN JUAN COUNTY BENCHMARK #00165, HAVING AN ELEVATION OF 34.28, PER LAND CORNER RECORD #2000 0303018.

ON-SITE SURVEY CONTROL TABLE

POINT	NORTHING	EASTING	ELEVATION	DESCRIPTION
205	629,477.19	1,139,257.92	23.74	REBAR AND PLASTIC CAP
206	629,622.13	1,139,287.90	22.40	REBAR AND PLASTIC CAP
208	629,620.32	1,139,364.78	23.16	PK NAIL AND WASHER
211	629,528.31	1,139,360.16	23.28	PK NAIL AND WASHER

SURVEYOR'S CERTIFICATE

I HEREBY CERTIFY THAT I AM A LICENSED LAND SURVEYOR IN THE STATE OF WASHINGTON, THAT THIS MAP IS BASED ON AN ACTUAL FIELD SURVEY DONE BY ME OR UNDER MY DIRECT SUPERVISION AND THAT ALL DATA SHOWN HEREON ACTUALLY EXISTS IN THE LOCATIONS SHOWN AT THE TIME OF THIS SURVEY. THIS TOPOGRAPHIC MAP WAS DONE AT THE REQUEST OF EASTSOUND SEWER AND WATER DISTRICT IN 2021.

PAUL JONATHAN DARROW, P.L.S. NO. 50697 DATE

LEGAL DESCRIPTION PER AFN. 2021-1022007

THE NORTH 144 FEET OF THE WEST 148 FEET OF THE SOUTH HALF OF THE SOUTHEAST QUARTER OF THE NORTHWEST QUARTER OF THE SOUTHEAST QUARTER OF SECTION 11, TOWNSHIP 37 NORTH, RANGE 2 WEST, W.M., IN SAN JUAN COUNTY, WASHINGTON.

TOGETHER WITH THE SOUTH 10 FEET OF THE NORTH 154 FEET OF THE EAST 112 FEET OF THE WEST 148 FEET OF THE SOUTH HALF OF THE SOUTHEAST QUARTER OF THE NORTHWEST QUARTER OF THE SOUTHEAST QUARTER OF SECTION 11, TOWNSHIP 37 NORTH, RANGE 2 WEST, W.M., IN SAN JUAN COUNTY, WASHINGTON.

RECORD DOCUMENTS

- BOOK 3, PAGE 61 - PLAT OF SURVEY FOR EASTSOUND WATER DISTRICT
- BOOK 18, PAGE 131 - RECORD OF SURVEY FOR PORT OF ORCAS
- BOOK 16, PAGE 99 - RECORD OF SURVEY FOR SAN JUAN COUNTY

SURVEYOR'S NOTES

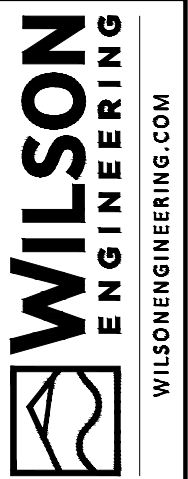
1. THIS TOPOGRAPHIC SURVEY WAS ORIGINALLY PERFORMED IN JANUARY 2013. ADDITIONAL DATA WAS SURVEYED IN MARCH 2021. ALL MONUMENTS SHOWN HEREON WERE VISITED DURING THE COURSE OF THIS SURVEY UNLESS OTHERWISE NOTED. THIS SURVEY CORRECTLY SHOWS THE AREA OF THE SUBJECT PROPERTY AND THE LOCATION OF ALL STRUCTURES AND IMPROVEMENTS SITUATED ON THE SUBJECT PROPERTY.

2. ANGULAR AND LINEAR MEASUREMENTS WERE COLLECTED USING A COMBINATION OF GPS AND CONVENTIONAL METHODOLOGIES. PRIMARY CONTROL WAS COLLECTED USING TRIMBLE 5700 SURVEY-GRADE GPS RECEIVERS OPERATING IN NETWORKED RTK MODE. FROM GPS CONTROL, A TRIMBLE S7 ROBOTIC TOTAL STATION WAS USED TO TIE SECONDARY CONTROL POINTS, COLLECT TOPOGRAPHIC DATA AND MEASURE LOT CORNERS.

3. THE LOCATION OF UTILITIES SHOWN HEREON ARE FROM OBSERVED EVIDENCE OF ABOVE GROUND APPURTENANCES TOGETHER WITH MARKINGS BY UTILITY COMPANIES.



BID SET



DESIGNED BY JCC
DRAWN BY SNR
CHECKED BY AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

W.A.C. 332-130 COMPLIANCE SHEET

DATE
NOV 2, 2022

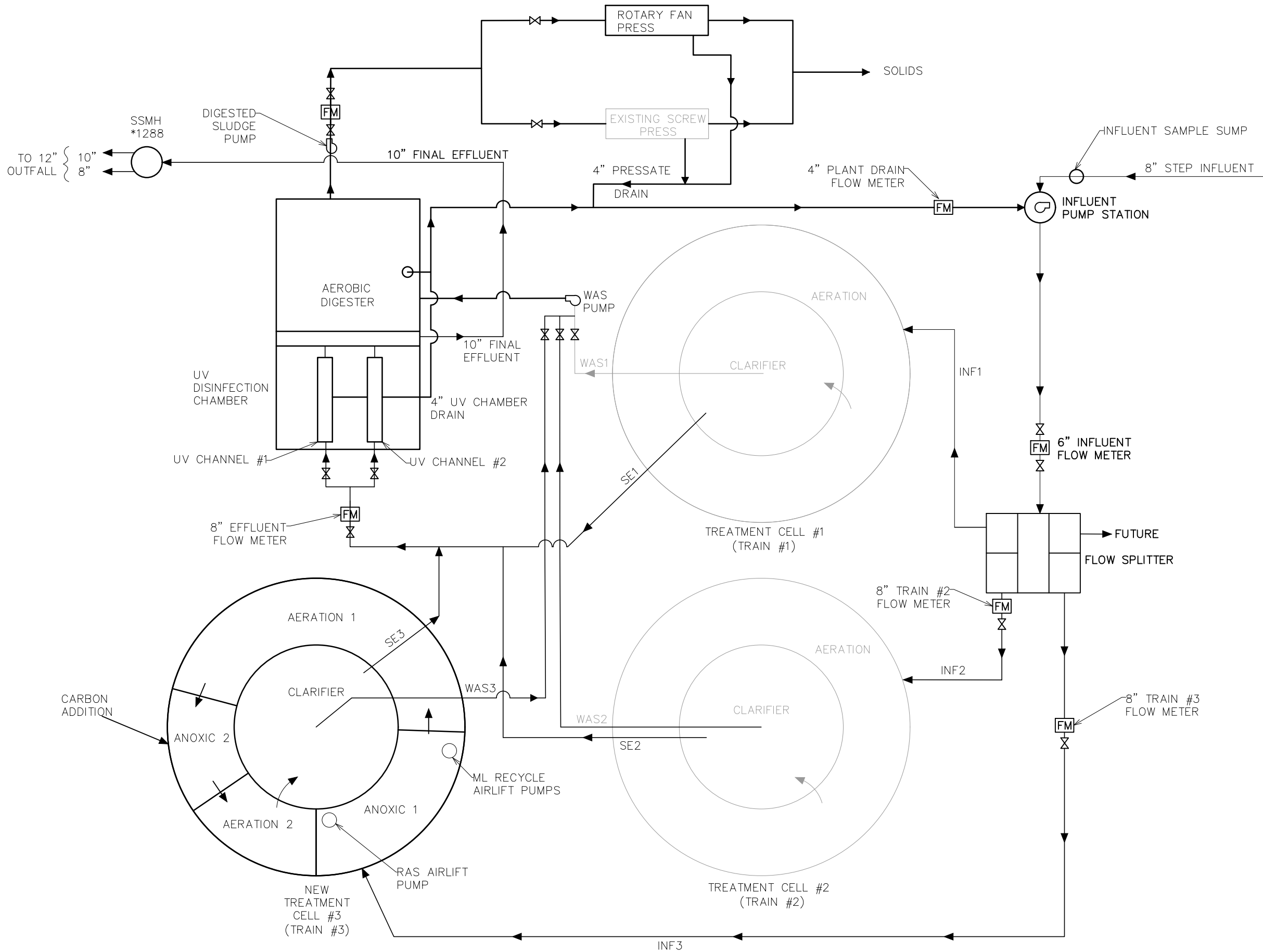
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LEGEND	
	ISOLATION VALVE
	FLOW DIRECTION
	FLOW METER
	AIRLIFT PUMP



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NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
PROCESS SCHEMATIC

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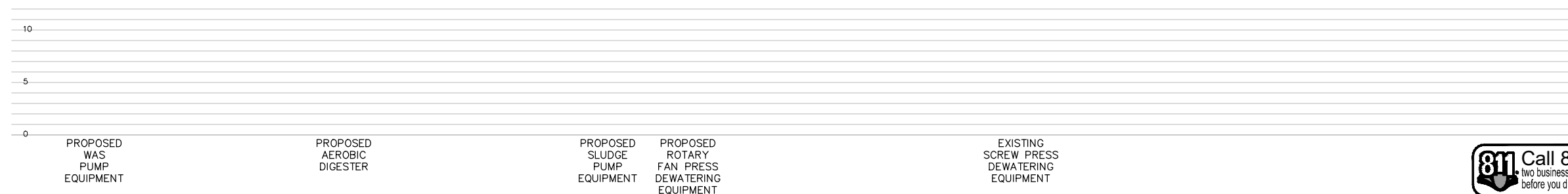
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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

HYDRAULIC PROFILE

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NO.	REVISIONS	BY	DATE
3	ADD UV DISINFECTION	JGC	5-10-22

WASTEWATER TREATMENT PLANT DESIGN INFORMATION

START-UP DESIGN INFORMATION
(AFTER PHASE I IS COMPLETED)

FLOW:

ADF	= 0.10 MGD
MMF	= 0.12 MGD
MAX. DAY	= 0.30 MGD
PEAK HR	= 0.49 MGD

AVERAGE DAILY LOADS:

BOD	= 189 mg/L
TSS	= 46 mg/L
Ammonia-N	= 47 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 47 mg/L
TKN	= 62 mg/L
Alkalinity	= 328 mg/L

MAX MONTH LOADS:

BOD	= 250 mg/L
TSS	= 59 mg/L
Ammonia-N	= 62 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 62 mg/L
TKN	= 82 mg/L
Alkalinity	= 433 mg/L

CLARIFIER INFORMATION:

NO. OF UNITS:	3
DIAMETER:	18 FT
SWD:	CELL 1: 13.25 FT, CELL 2: 13.25' FT, CELL 3: 15.50'
RAS:	0.5Q TO 1.0Q

CLARIFIERS

ADF SOR:	197 GAL / DAY / SF
MMF SOR:	236 GAL / DAY / SF
MAX DAY:	588 GAL / DAY / SF

AERATION BASIN INFORMATION:

NO. OF UNITS: 3

TRAIN #1 (EXISTING)

VOLUME:	0.054 MG
ADF HRT:	38.9 HOURS
MMF HRT:	32.4 HOURS
MAX DAY:	13.0 HOURS

TRAIN #2 (EXISTING)

VOLUME:	0.054 MG
ADF HRT:	38.9 HOURS
MMF HRT:	32.4 HOURS
MAX DAY:	13.0 HOURS

TRAIN #3 (PROPOSED)

VOLUME:	0.0397 MG
ADF HRT:	28.1 HOURS
MMF HRT:	23.3 HOURS
MAX DAY:	9.4 HOURS

ANOXIC BASIN INFORMATION

NO. OF UNITS: 1

TRAIN #4 (PROPOSED)

VOLUME:	0.0245 MG
ADF HRT:	18 HOURS
MMF HRT:	15 HOURS
MAX DAY:	6 HOURS

UV DISINFECTION

MONTHLY GEOMETRIC MEAN: 200 FC/100 ML
WEEKLY GEOMETRIC MEAN: 400 FC/100 ML

CAPACITY OF 1 CHANNEL: 0.5 MGD
CAPACITY OF 2 CHANNELS : 1.0 MGD

AEROBIC DIGESTER

VOL: 15,000 GAL
ADF STORAGE AT 2.0% SOLIDS = 4.5 WEEKS
MMF STORAGE AT 2.0% SOLIDS = 2.9 WEEKS
MMF STORAGE AT 1.0% SOLIDS = 1.5 WEEKS

BUILD-OUT DESIGN INFORMATION
(YR - 2040 AFTER PHASE II IS COMPLETED)

FLOW:

ADF	= 0.19 MGD
MMF	= 0.22 MGD
MAX. DAY	= 0.46 MGD
PEAK HR	= 0.77 MGD

AVERAGE DAILY LOADS:

BOD	= 181 mg/L
TSS	= 44 mg/L
Ammonia-N	= 45 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 45 mg/L
TKN	= 60 mg/L
Alkalinity	= 314 mg/L

MAX MONTH LOADS:

BOD	= 250 mg/L
TSS	= 59 mg/L
Ammonia-N	= 62 mg/L
Nitrate/Nitrite-N	= 0 mg/L
TIN	= 62 mg/L
TKN	= 82 mg/L
Alkalinity	= 433 mg/L

CLARIFIER INFORMATION:

NO. OF UNITS:	3
DIAMETER:	18 FT
SWD:	15.50 FT
RAS:	0.5Q TO 1.0Q

CLARIFIERS

ADF SOR:	249 GAL / DAY / SF
MMF SOR:	286 GAL / DAY / SF
MAX DAY:	603 GAL / DAY / SF
PEAK HR:	1008 GAL / DAY / SF

AERATION & ANOXIC BASIN INFORMATION:

NO. OF TRAINS: 3
NO. OF AERATION BASINS PER TRAIN: 2
NO. OF ANOXIC BASINS PER TRAIN: 2
TOTAL VOLUME OF AERATION & ANOXIC BASINS PER TRAIN: 0.0642 MG
MMF HRT: 21 HOURS

UV DISINFECTION

MONTHLY GEOMETRIC MEAN: 200 FC/100 ML
WEEKLY GEOMETRIC MEAN: 400 FC/100 ML

CAPACITY OF 1 CHANNEL: 0.5 MGD
CAPACITY OF 2 CHANNELS : 1.0 MGD

AEROBIC DIGESTER

VOL: 15,000 GAL
ADF STORAGE AT 2.0% SOLIDS (WITH CARBON ADDITION) = 1.9 WEEKS
MMF STORAGE AT 2.0% SOLIDS (WITH CARBON ADDITION) = 1.2 WEEKS

PROCESS DESIGN CRITERIA

TRAIN 3 AERATION BASIN: DESIGN INFORMATION

- AERATION BASIN MIXING = MEDIUM BUBBLE AERATION MIXING
- ANOXIC BASIN MIXER = SUBMERSIBLE MIXERS
 - ONE 5 HP SUBMERSIBLE MIXER IN ANOXIC ZONE #1: 15,686 GALLONS.
 - ONE 5 HP SUBMERSIBLE MIXER IN ANOXIC ZONE #2: 8,789 GALLONS.
- AERATION BASINS
 - AERATION ZONE #1: 29,793 GALLONS, 8 DIFFUSERS. DESIGN: 24 SCFM/DIFFUSER, 192 SCFM TOTAL
 - AERATION ZONE #2: 9,889 GALLONS, 2 DIFFUSERS. DESIGN: 24 SCFM/DIFFUSER, 48 SCFM TOTAL
 - TOTAL AERATION AIR = 240 SCFM
- AIR LIFT PUMPS
 - AIR NEEDED FOR RAS AIRLIFT PUMP = 15 SCFM. PIPE SIZE = 3-INCH. FLOW = 51 GPM. TARGET = 1Q
 - AIR NEEDED FOR NITRATE RECYCLE AIRLIFT PUMP = 61 SCFM. PIPE SIZE = 6-INCH. FLOW = 203 GPM. TARGET = 4Q
 - TOTAL AIRLIFT AIR = 76 SCFM
- TOTAL AIR REQUIRED = 316 SCFM AT 9.2 PSIG
- SRT = 15 DAYS (YR-2040 MMF)
- HRT = 21 HOURS (YR-2040 MMF)
- MLSS = 3000 MG/L (RANGE = 2000 TO 4000 MG/L)
- F/M = 0.10 TO 0.20 (LBS INFLUENT BOD/LBS MLVSS)
- VOLUMETRIC LOAD = 10 TO 30 (LBS. INFLUENT BOD/KCF DAY)

UV DISINFECTION: DESIGN INFORMATION

- UV DOSAGE = 30,000 MJ/CM2
- DESIGN UV TRANSMITTANCE = 50% OR GREATER
- LAMPS: 2 PER MODULE, 8 MODULES PER CHANNEL, 2 CHANNELS TOTAL
- HYDRAULIC CAPACITY = 0.5 MGD PER CHANNEL, TOTAL CAPACITY = 1.0 MGD

AEROBIC DIGESTER: DESIGN INFORMATION

- AERATION: COARSE BUBBLE. CAPACITY = 20 CFM/DIFFUSER X 6 DIFFUSERS = 120 CFM TOTAL.
- DIGESTER VOLUME = 15,000 GALLONS = 2,000 CF
- AIR NEEDED FOR MIXING = 30 CFM PER KCF OF STORAGE VOLUME = 30 X 2 = 60 CFM
- MLSS RANGE = 2000 TO 4000 MG/L
- WWTP MCRT RANGE = 5 TO 15 DAYS
- DIGESTER SOLIDS RETENTION TIME = 7 DAYS TO 28 DAYS (1 TO 4 WEEKS)
- DECANT CONTROL: MANUAL T-VALVE ADJUSTMENTS
- WEEKLY DECANT VOLUME = 50% TO 75% OF WEEKLY WAS VOLUME (WAS = 10,000 TO 18,000 GAL/WK; DECANT = 5000 TO 13,500 GAL/WK)

INFLUENT PUMP STATION: SIZE, TYPE, DESIGN CAPACITY

- 2 SMALL SUBMERSIBLE PUMPS. CAPACITY OF EACH PUMP = 120 TO 220 GPM
- 2 LARGE SUBMERSIBLE PUMPS. CAPACITY OF EACH PUMP = 320 TO 500 GPM
- TOTAL FIRM PS CAPACITY (WITH 1 SMALL AND 1 LARGE PUMP OUT OF SERVICE) = 535 GPM (0.77 MGD)

FLOW METER INFO: SIZE, TYPE, DESIGN CAPACITY

- INFLUENT: 6", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 10 FPS = 0 TO 880 GPM (1.27 MGD)
- TRAIN #2: 8", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 3 FPS = 0 TO 470 GPM (0.68 MGD)
- TRAIN #3: 8", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 3 FPS = 0 TO 470 GPM (0.68 MGD)
- EFFLUENT: 8", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 6 FPS = 0 TO 940 GPM (1.35 MGD)
- PLANT DRAIN: 4", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 5 FPS = 0 TO 200 GPM (0.29 MGD)
- 2W WATER: 1", ELECTROMAGNETIC FLOW METER, VELOCITY = 0 TO 20 FPS = 0 TO 25 GPM

RAS RECYCLE: DESIGN INFORMATION

AIR NEEDED FOR RAS AIRLIFT PUMP = 15 SCFM. PIPE SIZE = 3-INCH. FLOW = 51 GPM. TARGET = 1Q

NITRATE RECYCLE: DESIGN INFORMATION

AIR NEEDED FOR NITRATE RECYCLE AIRLIFT PUMP = 61 SCFM. PIPE SIZE = 6-INCH. FLOW = 203 GPM. TARGET = 4Q

WAS PUMPS: DESIGN INFORMATION

- NUMBER OF PUMPS: 1 DUTY PUMP, 1 STANDBY PUMP (SWING PUMP)
- PUMP STYLE: HOSE PUMP (PERISTALTIC)
- PUMP CAPACITY: 0 TO 40 GPM (ANTICIPATED SETPOINT RANGE 10 GPM TO 30 GPM)

DIGESTER PUMPS: DESIGN INFORMATION

- NUMBER OF PUMPS: 1 DUTY PUMP, 1 STANDBY PUMP (SWING PUMP)
- PUMP STYLE: HOSE PUMP (PERISTALTIC)
- PUMP CAPACITY: 0 TO 40 GPM (ANTICIPATED SETPOINT RANGE 12 GPM TO 20 GPM)

ROTARY FAN PRESS: DESIGN INFORMATION

- NUMBER OF DEWATERING UNITS: 1 UNIT
- DEWATERING UNIT STYLE: ROTARY FAN SYSTEM
- DEWATERING UNIT INPUT CAPACITY: 12 GPM AT 2% SOLIDS, 20 GPM AT 0.5% SOLIDS
- DEWATERING UNIT OUTPUT CAPACITY:
 - 12% TO 16% CAKE SOLIDS
 - WET CAKE THROUGHPUT: UP TO 1000 WET LBS. PER HOUR
 - DRY CAKE THROUGHPUT: UP TO 120 DRY LBS. PER HOUR

2W WATER SYSTEM PUMPS: DESIGN INFORMATION

- NUMBER OF PUMPS: 1 DUTY PUMP, 1 STANDBY PUMP
- PUMP STYLE: SUBMERSIBLE PUMP
- PUMP CAPACITY: 20 TO 80 GPM (VFD CONTROL)

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DESIGN CRITERIA

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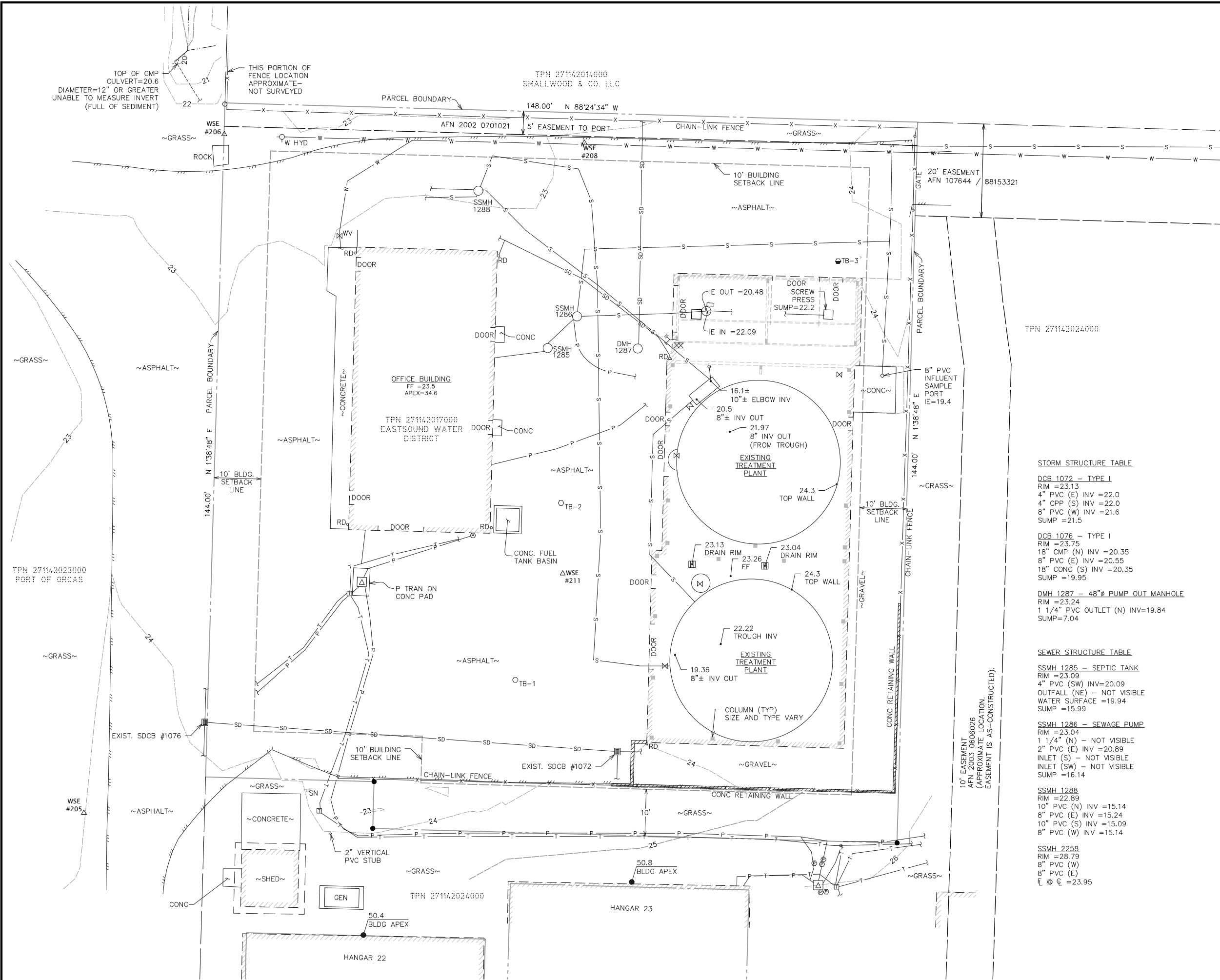
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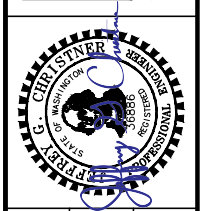
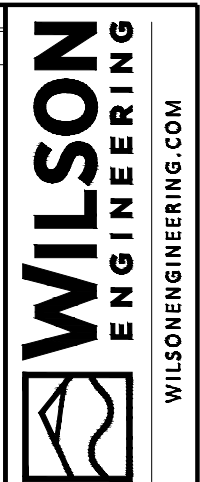
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SAN JUAN COUNTY	WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I	
EXISTING CONDITIONS	

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REQUIRED TESC ELEMENTS

ELEMENT 1: PRESERVE VEGETATION/MARK CLEARING LIMITS

A) BEFORE BEGINNING LAND DISTURBING ACTIVITIES, INCLUDING CLEARING AND GRADING, CLEARLY MARK ALL CLEARING LIMITS, SENSITIVE AREAS AND THEIR BUFFERS, AND TREES THAT ARE TO BE PRESERVED WITHIN THE CONSTRUCTION AREA.
B) RETAIN THE DUFF LAYER, NATIVE TOP SOIL, AND NATURAL VEGETATION IN AN UNDISTURBED STATE TO THE MAXIMUM DEGREE PRACTICABLE.

ELEMENT 2: ESTABLISH CONSTRUCTION ACCESS

A) LIMIT CONSTRUCTION VEHICLE ACCESS AND EXIT TO ONE ROUTE, IF POSSIBLE.
B) STABILIZE ACCESS POINTS WITH A PAD OF QUARRY SPALLS, CRUSHED ROCK, OR OTHER EQUIVALENT BMPS, TO MINIMIZE TRACKING OF SEDIMENT ONTO PUBLIC ROADS.
C) LOCATE WHEEL WASH OR TIRE BATHS ON SITE, IF THE STABILIZED CONSTRUCTION ENTRANCE IS NOT EFFECTIVE IN PREVENTING TRACKING SEDIMENT ONTO ROADS.
D) IF SEDIMENT IS TRACKED OFF SITE, CLEAN THE AFFECTED ROADWAY THOROUGHLY AT THE END OF EACH DAY, OR MORE FREQUENTLY AS NECESSARY (FOR EXAMPLE, DURING WET WEATHER). REMOVE SEDIMENT FROM ROADS BY SHOVELING, SWEEPING, OR PICK UP AND TRANSPORT THE SEDIMENT TO A CONTROLLED SEDIMENT DISPOSAL AREA.
E) CONDUCT STREET WASHING ONLY AFTER SEDIMENT IS REMOVED IN ACCORDANCE WITH THE ABOVE ITEM.
F) CONTROL STREET WASH WASTEWATER BY PUMPING BACK ON-SITE, OR OTHERWISE PREVENT IT FROM DISCHARGING INTO SYSTEMS TRIBUTARY TO WATERS OF THE STATE.

ELEMENT 3: CONTROL FLOW RATES

A) THIS ELEMENT IS NOT REQUIRED ON SITE THAT ARE THIS SMALL.

ELEMENT 4: INSTALL SEDIMENT CONTROLS

A) DESIGN, INSTALL, AND MAINTAIN EFFECTIVE EROSION CONTROLS AND SEDIMENT CONTROLS TO MINIMIZE THE DISCHARGE OF POLLUTANTS.
B) CONSTRUCT SEDIMENT CONTROL BMPS (SEDIMENT, TRAPS, SILT FENCE, INLET PROTECTION, ETC.) AS ONE OF THE FIRST STEPS IN GRADING. THESE BMPS SHALL BE FUNCTIONAL BEFORE OTHER LAND DISTURBING ACTIVITIES TAKE PLACE.
C) MINIMIZE SEDIMENT DISCHARGES FROM THE SITE. THE DESIGN, INSTALLATION AND MAINTENANCE OF EROSION AND SEDIMENT CONTROLS MUST ADDRESS FACTORS SUCH AS THE AMOUNT, FREQUENCY, INTENSITY AND DURATION OF PRECIPITATION, THE NATURE OF RESULTING STORMWATER RUNOFF, AND SOIL CHARACTERISTICS, INCLUDING THE RANGE OF SOIL PARTICLE SIZES EXPECTED TO BE PRESENT ON THE SITE.
D) DIRECT STORMWATER RUNOFF FROM DISTURBED AREAS THROUGH A SEDIMENT REMOVAL BMP, BEFORE THE RUNOFF LEAVES A CONSTRUCTION SITE. RUNOFF FROM FULLY STABILIZED AREAS MAY BE DISCHARGED WITHOUT A SEDIMENT REMOVAL BMP.
E) BMP C140 DUST CONTROL. CONTRACTOR SHALL KEEP DUST FROM CONSTRUCTION ACTIVITIES AND EXPOSED SOILS TO A MINIMUM.

ELEMENT 5: STABILIZE SOILS

A) STABILIZE EXPOSED AND UNWORKED SOILS BY APPLICATION OF EFFECTIVE BMPS THAT PREVENT EROSION. APPLICABLE BMPS INCLUDE, BUT ARE NOT LIMITED TO: TEMPORARY AND PERMANENT SEEDING, SODDING, MULCHING, PLASTIC COVERING, EROSION CONTROL FABRICS AND MATTING, SOIL APPLICATION OF POLYACRYLAMIDE (PAM), THE EARLY APPLICATION OF GRAVEL BASE EARLY ON AREAS TO BE PAVED, AND DUST CONTROL.
B) CONTROL STORMWATER VOLUME AND VELOCITY WITHIN THE SITE TO MINIMIZE SOIL EROSION.
C) SOILS MUST NOT REMAIN EXPOSED AND UNWORKED FOR MORE THAN THE TIME PERIODS SET FORTH BELOW TO PREVENT EROSION:
- DURING THE DRY SEASON (MAY 1 - SEPT. 30): 7 DAYS
- DURING THE WET SEASON (OCTOBER 1 - APRIL 30): 2 DAYS
D) STABILIZE SOILS AT THE END OF THE SHIFT BEFORE A HOLIDAY OR WEEKEND IF NEEDED BASED ON THE WEATHER FORECAST.
E) STABILIZE SOIL STOCKPILES FROM EROSION, PROTECTED WITH SEDIMENT TRAPPING MEASURES, AND WHERE POSSIBLE, BE LOCATED AWAY FROM STORM DRAIN INLETS, WATERWAYS AND DRAINAGE CHANNELS.
F) MINIMIZE THE AMOUNT OF SOIL EXPOSED DURING CONSTRUCTION ACTIVITY.
G) MINIMIZE THE DISTURBANCE OF STEEP SLOPES.
H) PRESERVE TOPSOIL AND MINIMIZE COMPACTING OF THESE AREAS.

ELEMENT 6: PROTECT SLOPES

A) DESIGN AND CONSTRUCT CUT-AND-FILL SLOPES IN A MANNER TO MINIMIZE EROSION. APPLICABLE PRACTICES INCLUDE, BUT ARE NOT LIMITED TO, REDUCING CONTINUOUS LENGTH OF SLOPE WITH TERRACING AND DIVERSIONS, REDUCING SLOPE STEEPNESS, AND ROUGHENING SLOPE SURFACES (FOR EXAMPLE, TRACK WALKING).
B) DIVERT OFF-SITE STORMWATER (RUN-ON) OR GROUND WATER AWAY FROM SLOPES AND DISTURBED AREAS WITH INTERCEPTOR DIKES, PIPES AND/OR SWALES. OFF-SITE STORMWATER SHOULD BE MANAGED SEPARATELY FROM STORMWATER GENERATED ON THE SITE.
C) AT THE TOP OF SLOPES, COLLECT DRAINAGE IN PIPE SLOPE DRAINS OR PROTECTED CHANNELS TO PREVENT EROSION.
D) PLACE EXCAVATED MATERIAL ON THE UPHILL SIDE OF TRENCHES, CONSISTENT WITH SAFETY AND SPACE CONSIDERATIONS.
E) PLACE CHECK DAMS AT REGULAR INTERVALS WITHIN CONSTRUCTED CHANNELS THAT ARE CUT DOWN A SLOPE.

ELEMENT 7: PROTECT DRAIN INLETS

A) PROTECT ALL STORM DRAIN INLETS MADE OPERABLE DURING CONSTRUCTION SO THAT STORMWATER RUNOFF SHALL NOT ENTER THE CONVEYANCE SYSTEM WITHOUT FIRST BEING FILTERED OR TREATED TO REMOVE SEDIMENT.
B) CLEAN OR REMOVE AND REPLACE INLET PROTECTION DEVICES WHEN SEDIMENT HAS FILLED ONE-THIRD OF THE AVAILABLE STORAGE (UNLESS A DIFFERENT STANDARD IS SPECIFIED BY THE PRODUCT MANUFACTURER).

ELEMENT 8: STABILIZE CHANNELS AND OUTLETS

A) DESIGN, CONSTRUCT, AND STABILIZE ALL ON-SITE CONVEYANCE CHANNELS TO PREVENT EROSION.
B) PROVIDE STABILIZATION, INCLUDING ARMORING MATERIAL, ADEQUATE TO PREVENT EROSION OF OUTLETS, ADJACENT STREAM BANKS, SLOPES AND DOWNSTREAM REACHES AT THE OUTLETS OF ALL CONVEYANCE SYSTEMS.

ELEMENT 9: CONTROL POLLUTANTS

A) DESIGN, INSTALL, IMPLEMENT AND MAINTAIN EFFECTIVE POLLUTION PREVENTION MEASURES TO MINIMIZE THE DISCHARGE OF POLLUTANTS.

B) HANDLE AND DISPOSE OF ALL POLLUTANTS, INCLUDING WASTE MATERIALS AND DEMOLITION DEBRIS THAT OCCUR ON-SITE IN A MANNER THAT DOES NOT CAUSE CONTAMINATION OF STORMWATER.
C) PROVIDE COVER, CONTAINMENT, AND PROTECTION FROM VANDALISM FOR ALL CHEMICALS, LIQUID PRODUCTS, PETROLEUM PRODUCTS, AND OTHER MATERIALS THAT HAVE THE POTENTIAL TO POSE A THREAT TO HUMAN HEALTH OR THE ENVIRONMENT. ON-SITE FUELING TANKS MUST INCLUDE SECONDARY CONTAINMENT. SECONDARY CONTAINMENT MEANS PLACING TANKS OR CONTAINERS WITHIN AN IMPERVIOUS STRUCTURE CAPABLE OF CONTAINING 110% OF THE VOLUME CONTAINED IN THE LARGEST TAKE WITHIN THE CONTAINMENT STRUCTURE. DOUBLE-WALLED TANKS DO NOT REQUIRE ADDITIONAL SECONDARY CONTAINMENT.
D) CONDUCT MAINTENANCE, FUELING, AND REPAIR OF HEAVY EQUIPMENT AND VEHICLES USING SPILL PREVENTION AND CONTROL MEASURES. CLEAN CONTAMINATED SURFACES IMMEDIATELY FOLLOWING ANY SPILL INCIDENT.
E) DISCHARGE WHEEL WASH OR TIRE BATH WASTEWATER TO A SEPARATE ON-SITE TREATMENT SYSTEM THAT PREVENTS DISCHARGE TO SURFACE WATER, SUCH AS CLOSED-LOOP RECIRCULATION OR UPLAND APPLICATION, OR TO THE SANITARY SEWER, WITH LOCAL SEWER DISTRICT APPROVAL.
F) APPLY FERTILIZERS AND PESTICIDES IN A MANNER AND AT APPLICATION RATES THAT WILL NOT RESULT IN LOSS OF CHEMICAL TO STORMWATER RUNOFF. FOLLOW MANUFACTURERS' LABEL REQUIREMENTS FOR APPLICATION RATES AND PROCEDURES.
G) USE BMPS TO PREVENT CONTAMINATION OF STORMWATER RUNOFF BY PH MODIFYING SOURCES. THE SOURCES FOR THIS CONTAMINATION INCLUDE, BUT ARE NOT LIMITED TO: BULK CEMENT, CEMENT KILN DUST, FLY ASH, NEW CONCRETE WASHING AND CURING WATERS, WASTE STREAMS GENERATED FROM CONCRETE GRINDING AND SAWING, EXPOSED AGGREGATE PROCESSES, DEWATERING CONCRETE VAULTS, CONCRETE PUMPING AND MIXER WASHOUT WATERS.
H) ADJUST THE PH OF STORMWATER IF NECESSARY TO PREVENT VIOLATIONS OF WATER QUALITY STANDARDS.
I) ASSURE THAT WASHOUT OF CONCRETE TRUCKS IS PERFORMED OFF-SITE OR IN DESIGNATED CONCRETE WASHOUT AREAS ONLY. DO NOT WASH OUT CONCRETE TRUCKS ONTO THE GROUND, OR INTO STORM DRAINS, OPEN DITCHES, STREETS, OR STREAMS. DO NOT DUMP EXCESS CONCRETE ON-SITE, EXCEPT IN DESIGNATED CONCRETE WASHOUT AREAS. CONCRETE SPILLAGE OR CONCRETE DISCHARGE TO SURFACE WATERS OF THE STATE IS PROHIBITED.
J) OBTAIN WRITTEN APPROVAL FROM ECOLOGY BEFORE USING CHEMICAL TREATMENT OTHER THAN CO2 OR DRY ICE TO ADJUST PH.
ELEMENT 10: CONTROL DE-WATERING
A) DISCHARGE FOUNDATION, VAULT, AND TRENCH DE-WATERING WATER, WHICH HAS SIMILAR CHARACTERISTICS TO STORMWATER RUNOFF AT THE SITE, INTO A SEDIMENT REMOVAL BMP BEFORE DISCHARGING OFFSITE.
B) CLEAN, NON-TURBID DE-WATERING WATER, SUCH AS WELL-POINT GROUND WATER, CAN BE DISCHARGED TO SYSTEMS TRIBUTARY TO, OR DIRECTLY INTO SURFACE WATERS OF THE STATE PROVIDED THE DE-WATERING FLOW DOES NOT CAUSE EROSION OR FLOODING OF RECEIVING WATERS. DO NOT ROUTE CLEAN DEWATERING WATER THROUGH STORMWATER SEDIMENT PONDS.
C) HANDLE HIGHLY TURBID OR OTHERWISE CONTAMINATED DEWATERING WATER SEPARATELY FROM STORMWATER.
D) OTHER TREATMENT OR DISPOSAL OPTIONS MAY INCLUDE:
- TRANSPORT OFF-SITE IN A VEHICLE, SUCH AS A VACUUM FLUSH TRUCK, FOR LEGAL DISPOSAL IN A MANNER THAT DOES NOT POLLUTE STATE WATERS.
- ECOLOGY-APPROVED ON-SITE CHEMICAL TREATMENT OR OTHER SUITABLE TREATMENT TECHNOLOGIES.
- SANITARY OR COMBINED SEWER DISCHARGE WITH LOCAL SEWER DISTRICT APPROVAL, IF THERE IS NO OTHER OPTION.
- USE OF A SETTLING TANKS WITH OUTFALL TO A DITCH OR SWALE FOR SMALL VOLUMES OF LOCALIZED DEWATERING.
ELEMENT 11: MAINTAIN BMPS
A) MAINTAIN AND REPAIR ALL TEMPORARY AND PERMANENT EROSION AND SEDIMENT CONTROL BMPS AS NEEDED TO ASSURE CONTINUED PERFORMANCE OF THEIR INTENDED FUNCTION IN ACCORDANCE WITH BMP SPECIFICATIONS.
B) REMOVE ALL TEMPORARY EROSION AND SEDIMENT CONTROL BMPS WITHIN 30 DAYS AFTER ACHIEVING FINAL SITE STABILIZATION OR AFTER THE TEMPORARY BMPS ARE NO LONGER NEEDED.
ELEMENT 12: MANAGE THE PROJECT
A) PHASE LAND DISTURBANCE TO THE MAXIMUM DEGREE PRACTICABLE AND TAKE INTO ACCOUNT SEASONAL WORK LIMITATIONS.
B) THIS PROJECT IS NOT REQUIRED TO OBTAIN A CONSTRUCTION STORMWATER GENERAL PERMIT.
C) PROJECTS THAT DISTURB ONE OR MORE ACRES MUST HAVE SITE INSPECTIONS CONDUCTED BY A CERTIFIED EROSION AND SEDIMENT CONTROL LEAD (CESCL). PROJECT SITES DISTURBING LESS THAN ONE ACRE MAY HAVE A CESCL OR A PERSON WITHOUT CESCL CERTIFICATION CONDUCT INSPECTIONS. BY THE INITIATION OF CONSTRUCTION, THE SWPPP MUST IDENTIFY THE CESCL OR INSPECTOR, WHO MUST BE PRESENT ON-SITE OR ON-CALL AT ALL TIMES.
D) THE CESCL OR INSPECTOR (PROJECT SITES LESS THAN ONE ACRE) MUST HAVE THE SKILLS TO ASSESS THE:
- SITE CONDITIONS AND CONSTRUCTION ACTIVITIES THAT COULD IMPACT THE QUALITY OF STORMWATER.
- EFFECTIVENESS OF EROSION AND SEDIMENT CONTROL MEASURES USED TO CONTROL THE QUALITY OF STORMWATER DISCHARGES.
E) THE CESCL OR INSPECTOR MUST EXAMINE STORMWATER VISUALLY FOR THE PRESENCE OF SUSPENDED SEDIMENT, TURBIDITY, DISCOLORATION, AND OIL SHEEN. THEY MUST EVALUATE THE EFFECTIVENESS OF BMPS AND DETERMINE IF IT IS NECESSARY TO INSTALL, MAINTAIN, OR REPAIR BMPS TO IMPROVE THE QUALITY OF STORMWATER DISCHARGES. BASED ON THE RESULTS OF THE INSPECTION, CONSTRUCTION SITE OPERATORS MUST CORRECT THE PROBLEMS IDENTIFIED BY:
- REVIEWING THE SWPPP FOR COMPLIANCE WITH THE 13 CONSTRUCTION SWPPP ELEMENTS AND MAKING APPROPRIATE REVISIONS WITHIN 7 DAYS OF THE INSPECTION.
- IMMEDIATELY BEGINNING THE PROCESS OF FULLY IMPLEMENTING AND MAINTAINING APPROPRIATE SOURCE CONTROL AND/OR TREATMENT BMPS AS SOON AS POSSIBLE, ADDRESSING THE PROBLEMS NOT LATER THAN WITHIN 10 DAYS OF THE INSPECTION. IF INSTALLATION OF NECESSARY TREATMENT BMPS IS NOT FEASIBLE WITHIN 10 DAYS, THE CONSTRUCTION SITE OPERATOR MAY REQUEST AN EXTENSION WITHIN THE INITIAL 10-DAY RESPONSE PERIOD.
- DOCUMENTING BMP IMPLEMENTATION AND MAINTENANCE IN THE SITE LOG BOOK (SITES LARGER THAN 1 ACRE).
F) THE CESCL OR INSPECTOR MUST INSPECT ALL AREAS DISTURBED BY CONSTRUCTION

ACTIVITIES, ALL BMPS, AND ALL STORMWATER DISCHARGE POINTS AT LEAST ONCE EVERY CALENDAR WEEK AND WITHIN 24 HOURS OF ANY DISCHARGE FROM THE SITE. (FOR PURPOSES OF THIS CONDITION, INDIVIDUAL DISCHARGE EVENTS THAT LAST MORE THAN ONE DAY DO NOT REQUIRE DAILY INSPECTIONS. FOR EXAMPLE, IF A STORMWATER POND DISCHARGES CONTINUOUSLY OVER THE COURSE OF A WEEK, ONLY ONE INSPECTION IS REQUIRED THAT WEEK.) THE CESCL OR INSPECTOR MAY REDUCE THE INSPECTION FREQUENCY FOR TEMPORARY STABILIZED, INACTIVE SITES TO ONCE EVERY CALENDAR MONTH.

ELEMENT 13: PROTECT LOW IMPACT DEVELOPMENT BMPS

A) THERE ARE NOT LID BMPS ON THIS SITE.

GENERAL NOTES

1. BMPS: BEST MANAGEMENT PRACTICES (BMPS) REFERRED TO ON THIS PLAN AND IN THESE NOTES SHALL BE CONSTRUCTED AND MAINTAINED AS DESCRIBED IN MOST RECENT VERSION OF THE DEPARTMENT OF ECOLOGY'S STORMWATER MANAGEMENT MANUAL FOR WESTERN WASHINGTON (SWMMWW) SECTION ON CONSTRUCTION STORMWATER POLLUTION PREVENTION.
2. EXTENT: THE EXTENT OF EROSION AND SEDIMENTATION CONTROL MEASURES IS DEPENDENT ON WEATHER CONDITIONS, SITE SLOPES, LENGTH OF TIME GROUND IS LEFT EXPOSED, AND THE AREA OF EXPOSED GROUND. THE CONTRACTOR SHALL AT ALL TIMES MINIMIZE THE RISK OF SITE EROSION BY CAREFUL SCHEDULING AND BY IMPLEMENTING AND MAINTAINING BMPS UNTIL THE SITE IS PERMANENTLY STABILIZED.
3. VEGETATION: EXISTING VEGETATION SHALL BE PRESERVED WHERE ATTAINABLE.
4. ADJACENT PROPERTIES: PROPERTIES ADJACENT TO THE PROJECT SHALL BE PROTECTED FROM SEDIMENT DEPOSITION.
5. REMOVAL OF BMPS: ALL TEMPORARY EROSION AND SEDIMENT CONTROL BMPS SHALL BE REMOVED WITHIN 30 DAYS AFTER FINAL SITE STABILIZATION IS ACHIEVED OR AFTER THE TEMPORARY BMPS ARE NO LONGER NEEDED. TRAPPED SEDIMENT SHALL BE REMOVED OR STABILIZED ON-SITE. DISTURBED SOIL AREAS RESULTING FROM REMOVAL SHALL BE PERMANENTLY STABILIZED.
6. OTHER REQUIREMENTS: THE ENGINEER, OWNER, DEPARTMENT OF ECOLOGY, OR OTHER AGENCIES MAY REQUIRE BMPS IN ADDITION TO WHAT IS SHOWN ON THIS PLAN IF NECESSARY TO PREVENT VIOLATIONS OF SURFACE WATER QUALITY. THE CONTRACTOR SHALL IMPLEMENT THE BMPS AS REQUIRED.
7. THE IMPLEMENTATION OF THESE TESC PLANS AND THE CONSTRUCTION, MAINTENANCE, REPLACEMENT, AND UPGRADING OF THESE TESC FACILITIES IS THE RESPONSIBILITY OF THE CONTRACTOR UNTIL ALL CONSTRUCTION IS COMPLETED AND APPROVED AND VEGETATION/FINAL SURFACING IS ESTABLISHED.
8. THE TESC FACILITIES SHOWN ON THIS PLAN MUST BE CONSTRUCTED IN CONJUNCTION WITH ALL CLEARING AND GRADING ACTIVITIES, AND IN SUCH A MANNER AS TO INSURE THAT SEDIMENT AND SEDIMENT LADEN WATER DO NOT ENTER THE DRAINAGE SYSTEM, ROADWAYS, OR VIOLATE APPLICABLE WATER STANDARDS.
9. THE TESC FACILITIES SHOWN ON THIS PLAN ARE THE MINIMUM REQUIREMENTS FOR ANTICIPATED SITE CONDITIONS. DURING THE CONSTRUCTION PERIOD, THESE ESC FACILITIES SHALL BE UPGRADED AS NEEDED FOR UNEXPECTED STORM EVENTS AND TO ENSURE THAT SEDIMENT AND SEDIMENT LADEN WATER DO NOT LEAVE THE SITE.
10. OBTAIN PERMISSION AND/OR PERMIT APPROVAL AS REQUIRED FROM EASTSOUND WATER AND SEWER DISTRICT PRIOR TO ANY DISCHARGES TO THE SANITARY SEWER SYSTEM.
11. CONVEYANCE BYPASS: IF REQUIRED, PROVISION FOR BYPASS OF STORMWATER CONVEYANCE SHALL BE PROVIDED. BYPASS SHALL BE INSTALLED WHILE WORK IS IN PROGRESS AT A PARTICULAR LOCATION, BUT MATERIALS AND EQUIPMENT FOR IMMEDIATE INSTALLATION SHALL BE ON HAND. BYPASS SHALL BE IN PLACE WHILE SITE IS UNATTENDED FOR GREATER THAN 12 HOURS.
12. PUBLIC RIGHTS-OF WAY SHALL BE KEPT IN A CLEAN AND SERVICEABLE CONDITION AT ALL TIMES. IN THE EVENT MATERIALS ARE INADVERTENTLY DEPOSITED ON ROADWAYS, THE MATERIAL SHALL BE PROMPTLY REMOVED. MATERIALS ARE TO BE SWEEP AND REMOVED PRIOR TO ANY STREET FLUSHING. PUBLIC AND PRIVATE DRAINAGE WAYS SHALL BE PROTECTED FROM POLLUTION. NO MATERIAL IS TO BE DISCHARGED TO OR DEPOSITED IN STORMWATER SYSTEMS THAT MAY RESULT IN VIOLATION OF STATE OR FEDERAL WATER QUALITY STANDARDS.
13. ALL CATCH BASINS WITHIN THE PROJECT LIMITS SHALL BE CLEANED OUT AT THE COMPLETION OF THE PROJECT AND ANY MATERIAL REMOVED SHALL BE PROPERLY DISPOSED OF.
14. CONCRETE PROCESS WATER AND SLURRY FROM PAVEMENT CUTTINGS SHALL NOT BE ALLOWED TO ENTER ANY DRAINAGE CONVEYANCE. SLURRY/CUTTINGS SHALL NOT REMAIN ON PAVEMENT OVERNIGHT. ALL SAWCUTTING AND SURFACING OPERATIONS SHALL USE BMP C152 FOR POLLUTION PREVENTION.
15. PROJECT DESCRIPTION
a.TOTAL PROJECT AREA. 0.45 ACRES
b.TOTAL PROPOSED IMPERVIOUS AREA. 0.04 ACRES
c.TOTAL PROPOSED AREA TO BE DISTURBED. 0.30 ACRES
d.TOTAL VOLUME OF PROPOSED CUT/FILL. CUT = 3382 CY, FILL = 1598 CY. (ESTIMATED VOLUME BETWEEN EXISTING GROUND SURFACE AND PROPOSED FINISH GRADES FOR BASE BID AND ALL ALTERNATES. NO SHRINK OR SWELL FACTORS APPLIED.) VOLUMES ARE FOR TESC NARRATIVE ONLY AND ARE NOT INTENDED FOR BIDDING OR TAKE-OFF PURPOSES.
16. EXISTING SITE CONDITIONS
a.DESCRPTION OF EXISTING TOPOGRAPHY. EXISTING SLOPES VARY SLIGHTLY THROUGHOUT THE PROJECT SITE. MOST AREAS ARE FLAT TO SLOPED UP TO 2 PERCENT.
b.DESCRPTION OF EXISTING VEGETATION. AREAS ALONG THE NORTH AND SOUTH PROPERTY LINES ARE GRASS TURF. THERE ARE NO TREES OR SHRUBS ON THE SITE.
17. DESCRIPTION OF EXISTING DRAINAGE. RUNOFF FROM THE SITE FLOWS GENERALLY FROM SOUTHEAST TO NORTHWEST TO WHERE IT ENTERS A DITCH ON THE ADJOINING SITE ALONG THE NORTH BOUNDARY.

18. ADJACENT AREAS

a.DESCRPTION OF ADJACENT AREAS WHICH MAY BE AFFECTED BY SITE DISTURBANCE:
I.STREAMS. NONE.
II. LAKES. NONE.
III. WETLANDS. NONE.
IV. RESIDENTIAL AREAS. NONE.
V. ROADS. CESSNA ROAD, UNNAMED AIRPORT ACCESS ROADS.

b.DESCRPTION OF THE DOWNSTREAM DRAINAGE PATH LEADING FROM THE SITE TO THE RECEIVING BODY OF WATER (MINIMUM DISTANCE OF 400 YARDS). RUNOFF FROM THE SITE WILL CONTINUE WITH THE SAME PATTERN THAT EXISTS. RUNOFF WILL BE ROUTED TO A DITCH THAT IS JUST NORTHWEST OF THE SITE. RUNOFF IN THE DITCH IS CONVEYED NORTH THROUGH A SERIES OF DITCHES AND CULVERTS THROUGH PORT PROPERTY. IT APPEARS THAT THE OUTFALL MAY BE TO PUGET SOUND FURTHER TO THE NORTH.

19. CRITICAL AREAS

a.DESCRPTION OF CRITICAL AREAS THAT ARE ON OR ADJACENT TO THE SITE. THERE ARE NO CRITICAL AREAS WITHIN THE PROJECT SITE. THE SITE IS SURROUNDED BY DEVELOPMENT ON ALL SIDES. IT IS UNCLEAR IF THERE ARE CRITICAL AREAS NEAR THE SITE.
b.DESCRPTION OF SPECIAL REQUIREMENTS FOR WORKING IN OR NEAR CRITICAL AREAS. ALL WORK ON THIS PROJECT WILL BE DONE IN ACCORDANCE WITH ALL APPLICABLE LOCAL, STATE, AND FEDERAL PERMITS AS APPLICABLE. NO KNOWN CRITICAL AREAS ARE WITHIN OR ADJACENT TO THE PROJECT LIMITS.

20. DESCRIPTION OF ONSITE SOILS

a.THE FOLLOWING SOIL DESCRIPTIONS WERE TAKEN FROM THE USDA NRCS WEB SOIL SURVEY (HTTP://WEBSOILSURVEY.NRCS.USDA.GOV/APP/). SOILS IN THE WORK AREAS ARE CLASSIFIED AS:
-ROCHE-KILLEBREW COMPLEX. 2 TO 10 PERCENT SLOPES.

21. EROSION PROBLEM AREAS.

NO EROSION PROBLEM AREAS ARE EXPECTED.

22. CONSTRUCTION PHASING.

a.CONSTRUCTION SEQUENCE. THE CONSTRUCTION SEQUENCE WILL GENERALLY BE AS FOLLOWS;
• INSTALL EROSION PREVENTION FACILITIES: SILT FENCE, CONSTRUCTION ENTRANCE, INLET PROTECTION;
• REMOVE PORTIONS OF THE PAVEMENT TO INSTALL UNDERGROUND AND ABOVE GROUND SEWER STRUCTURES. CLEAR AND GRADE AREAS FOR BUILDING ADDITION AND RETAINING WALL.
• INSTALL ABOVE AND BELOW GROUND SEWER STRUCTURES, BUILDING ADDITION, AND RETAINING WALL
• REMOVE REMAINDER OF PAVEMENT AND INSTALL STORMWATER CONVEYANCE FACILITIES
• CLEAR NEW PAVING AREAS
• PAVE SITE WITH ASPHALT AND PROVIDE FINAL STABILIZATION TO LANDSCAPE AREAS.
b.CONSTRUCTION PHASING. CONTRACTOR IS ENCOURAGED TO PHASE WORK SUCH THAT DISTURBED AREAS ARE OPENED UP, IMPROVEMENTS INSTALLED, AND PERMANENTLY STABILIZED IN SECTIONS SUCH TO MINIMIZE POTENTIAL FOR EROSION.

23. CONSTRUCTION SCHEDULE

a.PROVIDE A PROPOSED CONSTRUCTION SCHEDULE. CONSTRUCTION IS EXPECTED TO OCCUR STARTING IN SPRING OF 2023 AND ENDING IN FALL OF 2024.
b.WET SEASON CONSTRUCTION ACTIVITIES. WET SEASON CONSTRUCTION ACTIVITIES ARE EXPECTED. INCREASED DILIGENCE WILL EMPLOYED TO PREVENT EROSION AND PROTECT DOWNSTREAM WATERS AND PROPERTIES FROM SEDIMENT. ALL WET SEASON HEIGHTENED TESC ACTIVITIES WILL BE IMPLEMENTED IN ACCORDANCE WITH THE SWMMWW.
I.PROPOSED WET SEASON CONSTRUCTION RESTRAINTS FOR ENVIRONMENTALLY SENSITIVE/CRITICAL AREAS. THERE ARE NO KNOWN ENVIRONMENTALLY SENSITIVE /CRITICAL AREAS WITHIN THE SITE, SO THERE ARE NO ADDITIONAL MEASURES TO BE UTILIZED.

24. ENGINEERING CALCULATIONS

a.SEDIMENT PONDS/TRAPS. NONE USED.
b.DIVERSIONS. NONE ARE ANTICIPATED
c.WATERWAYS. NONE USED.
d.RUNOFF/STORMWATER DETENTION CALCULATIONS. NONE NEEDED.



BID SET

WILSON
ENGINEERING





DESIGNED BY
JCC

DRAWN BY
JCS

CHECKED BY
AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DATE
NOV 2, 2022

SCALE
AS SHOWN

SHEET
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PAGE
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OF
96

JOB NUMBER
2020-070

T.E.S.C. NARRATIVE

11-02-2022

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LEGEND

	= STABILIZED CONSTRUCTION ENTRANCE PER $\frac{4}{C0.4}$
	= EMERGENCY ACCESS
	= SILT FENCE PER $\frac{3}{C0.4}$
	IP = INLET PROTECTION PER $\frac{2}{C0.4}$
	CD = CHECK DAM PER $\frac{5}{C0.4}$

KEYED NOTES

1 = EMERGENCY ACCESS. NO PARKING, BLOCKING, STOCKPIILING, STAGING ALLOWED. THIS AREA CAN BE USED FOR VEHICLE TURN AROUND

2 = SOLE INGRESS & EGRESS LOCATION

DCB 1072 - TYPE I
RIM = 23.13
4" PVC (E) INV = 22.0
4" CPP (S) INV = 22.0
8" PVC (W) INV = 21.6
SUMP = 21.5

DCB 1076 - TYPE I
RIM =23.75
18" CMP (N) INV =20.35
8" PVC (E) INV =20.55
18" CONC (S) INV =20.35
SUMP =19.95

DMH 1287 - 48"Ø PUMP OUT MANHOLE
RIM =23.24
1 1/4" PVC OUTLET (N) INV=19.84
SUMP=7.04

SSMH 1285 - SEPTIC TANK
RIM =23.09
4" PVC (SW) INV=20.09
OUTFALL (NE) - NOT VISIBLE
WATER SURFACE =19.94
SUMP =15.99

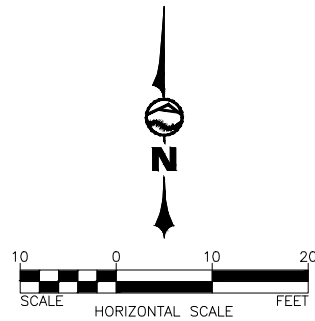
SSMH 1286 - SEWAGE PUMP
RIM = 23.04
1 1/4" (N) - NOT VISIBLE
2" PVC (E) INV = 20.89
INLET (S) - NOT VISIBLE
INLET (SW) - NOT VISIBLE
SUMP = 16.14

SSMH 1288
RIM = 22.89
10" PVC (N) INV = 15.14
8" PVC (E) INV = 15.24
10" PVC (S) INV = 15.09
8" PVC (W) INV = 15.14

SSMH 2258
RIM = 28.79
8" PVC (W)
8" PVC (E)
L @ C = 23.95

10' EASEMENT
AFN 2003 0606026
(APPROXIMATE LOCAL
EASEMENT IS AS-CO

1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.



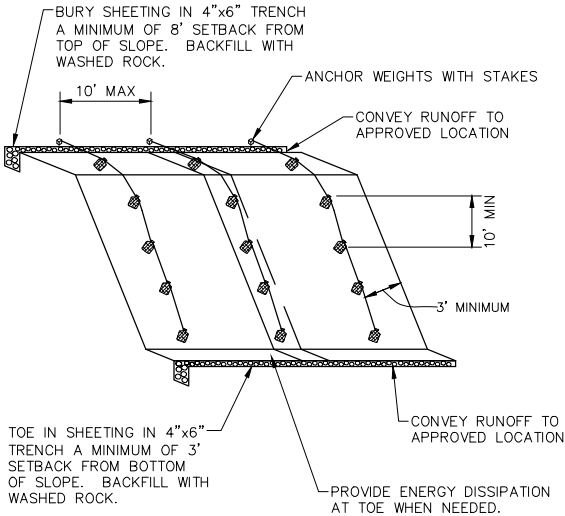
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NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

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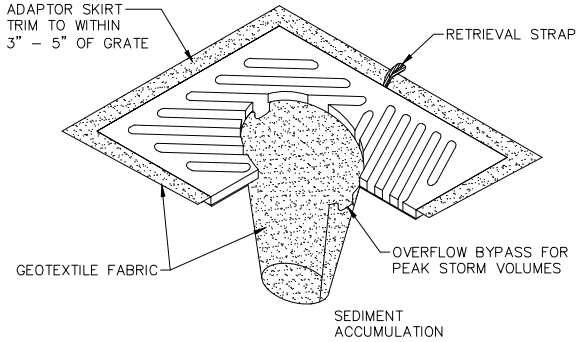
NOTES:

1. TIRES, SANDBAGS, OR EQUIVALENT MAY BE USED TO WEIGH DOWN PLASTIC SHEETING.
2. SEAMS BETWEEN SHEETS MUST OVERLAP A MINIMUM OF 12" AND BE WEIGHTED OR TAPED.
3. PLASTIC SHEETING SHALL HAVE A MINIMUM THICKNESS OF 6 MIL.
4. DUE TO RAPID RUNOFF CAUSED BY PLASTIC SHEETING, THIS METHOD SHALL NOT BE USED UPSLOPE OF AREAS THAT MIGHT BE ADVERSELY IMPACTED BY CONCENTRATED RUNOFF.
5. CONSTRUCT BERM OR SWALE AT TOP OF SLOPE AS DIRECTED BY THE CLEARING AND GRADING INSPECTOR.
6. CONSTRUCT DITCH AT BASE OF SLOPE IF NEEDED, AND DISCHARGE TO APPROVED LOCATION.

PLASTIC COVERING FOR SLOPES & STOCKPILES

1

NOT TO SCALE



NOTES:

1. INSERT SHALL BE INSTALLED PRIOR TO CLEARING AND GRADING ACTIVITY, OR UPON PLACEMENT OF A NEW CATCH BASIN.
2. SEDIMENT SHALL BE REMOVED FROM THE UNIT WHEN IT BECOMES HALF FULL .
3. SEDIMENT REMOVAL SHALL BE ACCOMPLISHED BY REMOVING THE INSERT, EMPTYING, AND RE-INSERTING IT INTO THE CATCH BASIN.

2

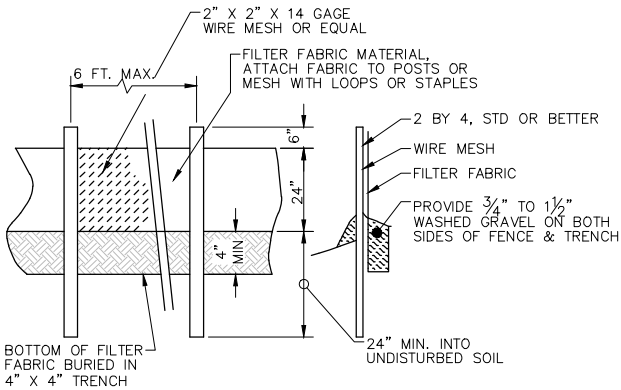
CATCH BASIN INSERT

NOT TO SCALE

NOTE: BASED ON FIGURE II-3.1: "STABILIZED CONSTRUCTION ACCESS" IN THE 2019 STORMWATER MANAGEMENT MANUAL FOR WESTERN WASHINGTON, REVISED JULY 2019, VOL II - CHAPTER 3 - PAGE 279

NOTES:

1. DRIVEWAY SHALL MEET THE REQUIREMENTS OF THE PERMITTING AGENCY.
2. IT IS RECOMMENDED THAT THE ACCESS BE CROWNED SO THAT RUNOFF DRAINS OFF THE PAD.

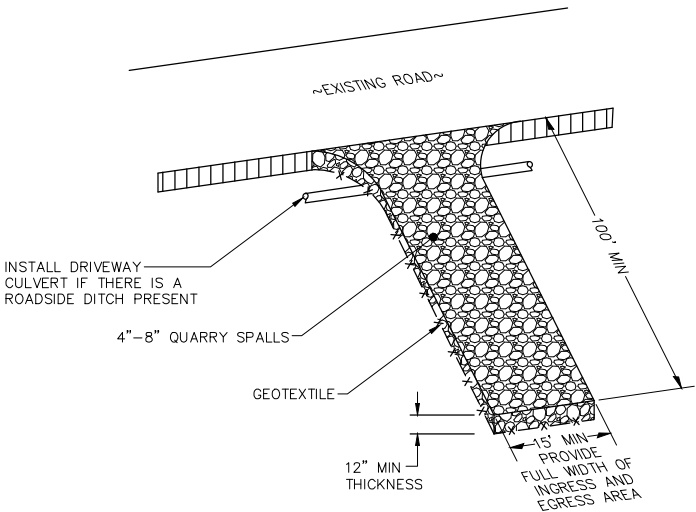


NOTE: SEE WSDOT STANDARD PLAN I-30.10-02 OR I-30.15-02, FOR HIGH VISIBILITY FENCE, SEE WSDOT STANDARD PLAN I-30.16-01 OR I-30.17-01.

3

FILTER FABRIC FENCE

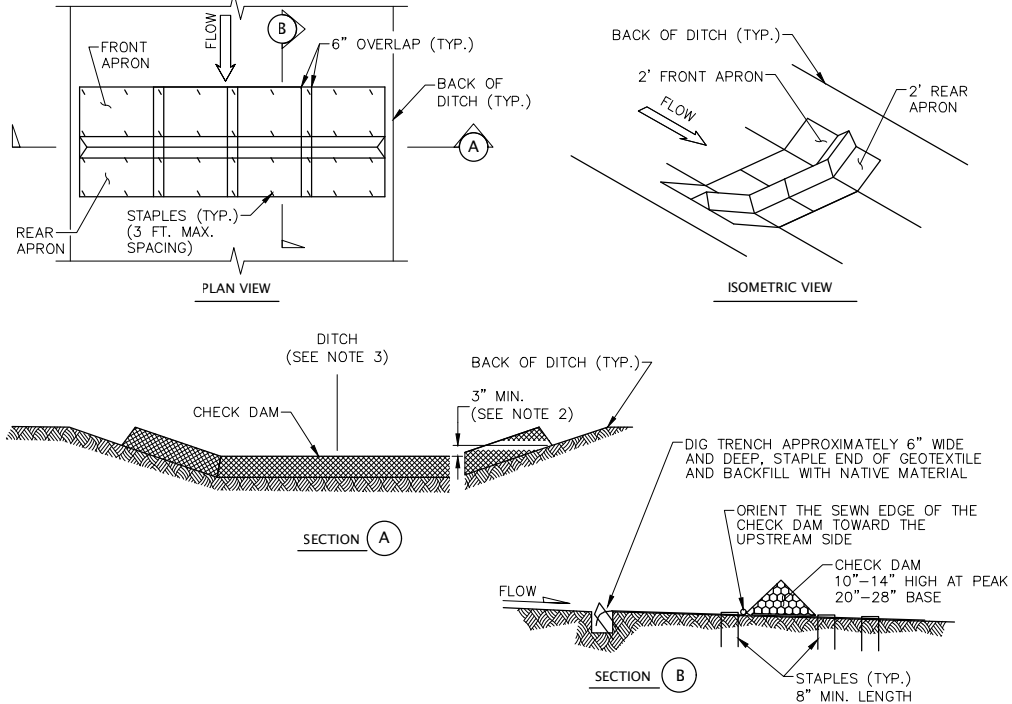
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4

STABILIZED CONSTRUCTION ENTRANCE

NOT TO SCALE



NOTES:

1. GEOTEXTILE ENCASED CHECK DAMS SHALL MEET THE REQUIREMENTS OF STANDARD SPECIFICATIONS 8-01.3(6)A AND 9-14.5(4).
2. INSTALL THE SLOPED ENDS OF THE CHECK DAM A MINIMUM OF 3" HIGHER THAN THE TOP OF THE CHECK DAM IN THE CHANNEL TO ENSURE THAT WATER FLOWS OVER THE DAM AND NOT AROUND IT.
3. FLAT BOTTOM DITCH DESIGN SHOWN, CHECK DAM INSTALLATION DETAILS ARE SIMILAR FOR "V" BOTTOM DITCHES.
4. PERFORM MAINTENANCE IN ACCORDANCE WITH STANDARD SPECIFICATION 8-01.3(15).

NOTE:

THE CONTRACTOR SHALL BE RESPONSIBLE FOR UPDATING COVERAGE UNDER THE DOE'S "CONSTRUCTION STORMWATER GENERAL PERMIT". CONTACT MELINDA WILSON, PERMIT ADMINISTRATOR AT DOE (360-870-8290), EMAIL: MELINDA.WILSON@ECY.WA.GOV) FOR ADDITIONAL INFORMATION.

5

GEOTEXTILE ENCASED CHECK DAM

NOT TO SCALE



DESIGNED BY

JGC

DRAWN BY

JGS

CHECKED BY

AWL

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WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

SAN JUAN COUNTY

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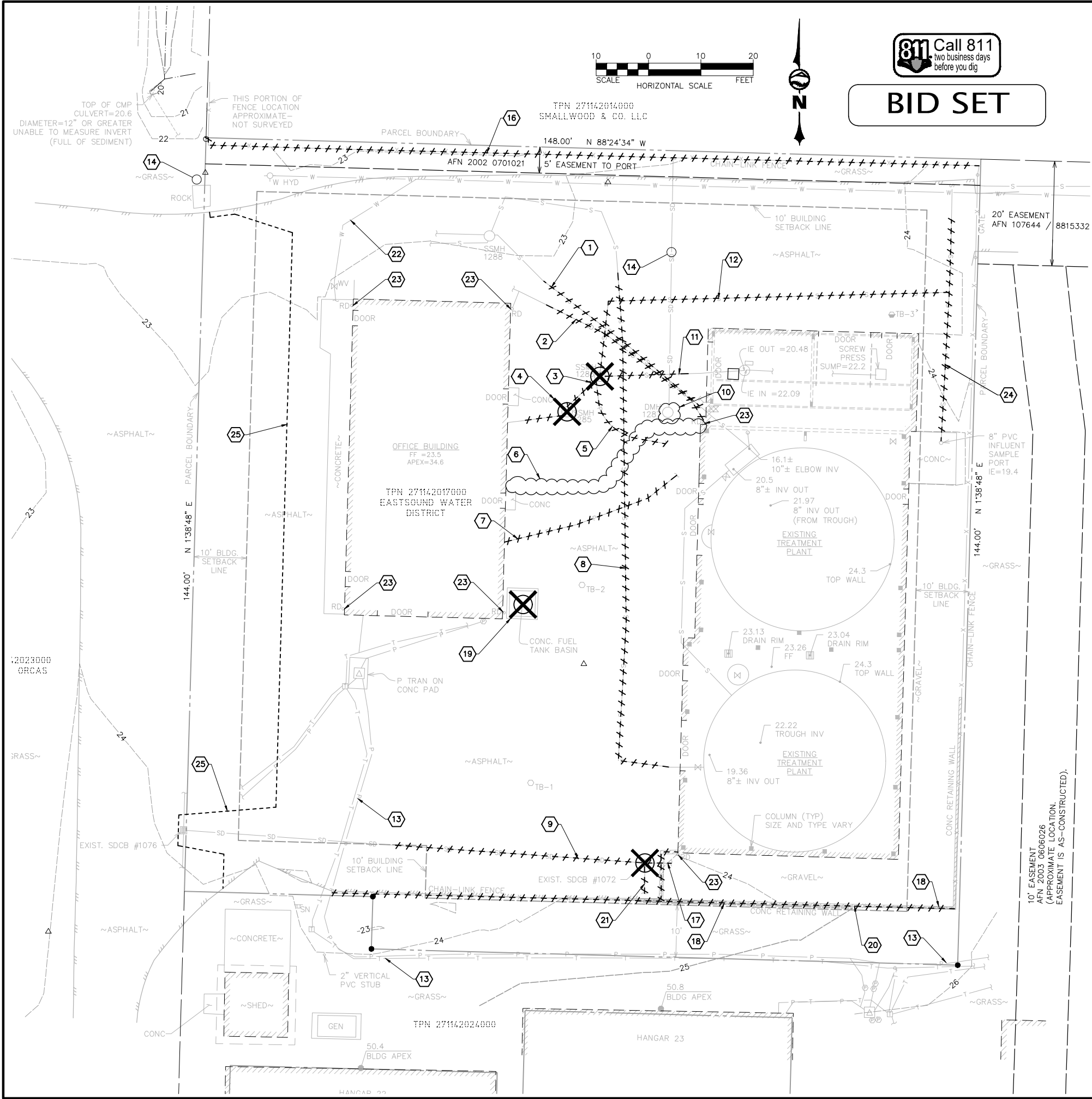
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6

ECOLOGY REVIEW COMMENTS

JGC

8/9/22

KEYED NOTES:

1

ABANDON AND REMOVE 40 LF OF 8" EFFLUENT (OUTFALL) PIPE AFTER INTERIM 8" HDPE PIPE IS INSTALLED AND TESTED. SEE SHEET C0.7 INTERIM PLAN OF OPERATION

2

ABANDON AND REMOVE 40 LF OF 4" ROOF DRAIN PIPE AFTER NEW 4" ROOF DRAIN GRAVITY PIPE IS INSTALLED AND TESTED. SEE C2 STORM DRAINAGE SHEETS

3

REMOVE EXISTING SUMP PUMP STATION AFTER NEW PLANT DRAIN PUMP STATION IS INSTALLED AND TESTED (OR PROVIDE TEMPORARY PUMPS, PIPING, AND BASINS AS NEEDED)

4

REMOVE EXISTING SEPTIC TANK AFTER NEW SEPTIC TANK AND INFLUENT P.S. ARE INSTALLED AND TESTED. PROVIDE TEMPORARY PUMPS AND BASINS AS NEEDED

5

MAINTAIN POWER TO EXISTING SUMP P.S. UNTIL NEW PLANT DRAIN P.S. IS INSTALLED AND TESTED

6

MAINTAIN EXISTING 2" WATER SERVICE UNTIL NEW WATER SERVICE IS INSTALLED AND TESTED. SEE C1 SITE PIPING SHEETS

7

RE-ROUTE ELECTRICAL AS NEEDED. SEE ELECTRICAL SHEETS

8

ABANDON AND REMOVE 60 LF OF 10" EFFLUENT (OUTFALL) PIPE AFTER INTERIM 8" HDPE PIPE IS INSTALLED AND TESTED. PROVIDE TEMPORARY RESTRAINED PLUG AT PIPE OPENING UNTIL NEW 10" EFFLUENT IS CONNECTED.

9

REMOVE EXISTING 87 LF STORM DRAIN & CATCH BASIN

10

PROTECT EXISTING STORM DRAIN PUMP STATION MANHOLE AND DISCHARGE PIPING

11

ABANDON AND REMOVE 20 LF OF 6" DRAIN PIPE TO SUMP PUMP STATION AFTER NEW PLANT DRAIN P.S. IS INSTALLED AND TESTED

12

ABANDON AND REMOVE 60 LF OF 13" SUMP PUMP FORCE MAIN PIPE, AFTER NEW PLANT DRAIN P.S. IS INSTALLED AND TESTED. PLUG CONNECTION AT 8" INFLUENT GRAVITY SEWER PIPE (EAST SIDE). SEE C1 SITE PIPING SHEETS

13

POTHOLE POWER & TELECOMMUNICATIONS PRIOR TO INSTALLING STORM SEWER OR RETAINING WALL OR FENCE. IF IN CONFLICT, RELOCATE AS NEEDED TO BE CLEAR OF PROPOSED FACILITIES. BED & BACKFILL SIMILAR TO STORM SEWER PIPE MIN. RELOCATED FACILITIES TO BE INSTALLED W/ 2' OF COVER MIN. ADDITIONAL COVER MAY BE REQUIRED TO AVOID CONFLICT WITH OTHER FACILITIES

14

POTHOLE EXISTING STORM SEWER AND REPORT PIPE LOCATION & ELEVATION TO THE ENGINEER PRIOR TO GRADING THE SITE OR INSTALLING ANY STORM SEWER FACILITIES.

15

EXISTING HOLDING TANKS TO BE REMOVED DURING CONSTRUCTION & RELOCATED ON SITE PER OWNER

16

REMOVE 149 LF OF CHAIN LINK FENCE

17

REMOVE 7' OF STORM DRAIN CONNECTED TO ROOF DRAIN

18

REMOVE 61 LF OF EXISTING RETAINING WALL. MAKE CLEAN CUT AT EAST END SUITABLE FOR MATCHING TO PROPOSED BLOCK WALL

19

EXISTING GENERATOR FUEL TANK AND CONCRETE FUEL SPILL BASIN TO BE REMOVED AND RESTORED AS ASPHALT PAVEMENT.

20

REMOVE 119 LF OF CHAIN LINK FENCE

21

REMOVE 8 LF OF 4" STORM DRAIN

22

RELOCATE WATER TO BE 1' CLEAR BELOW BOTTOM OF STORM DRAIN

23

DISCONNECT ROOF DRAINS AND PREPARE FOR CONNECTION TO PROPOSED DRAINS

24

ABANDON & REMOVE EXISTING 8" INFLUENT PIPE PER CONSTRUCTION SEQUENCE NOTES (SHEET C3.1)

25

SAWCUT ASPHALT PAVEMENT

CONSTRUCTION PHASING:

NOTE: ALL CONSTRUCTION PHASING STEPS SHALL BE COORDINATED WITH THE OWNER AND ENGINEER.

1.PERFORM DEMOLITION ITEMS 1 & 2.

2.LEAVING EXISTING INTERNAL PIPING IN PLACE AND OPERATIONAL, INSTALL NEW CONCRETE WALLS, WEIR GATES, AND IMPROVEMENTS TO EXISTING INTERIOR WALLS.

3.TIME CRITICAL STEP: PLUG OUTLET PIPE AT UPSTREAM MANHOLE AND BYPASS ALL FLOW. CLOSE INFLUENT VALVES TO PLANTS 1 & 2. THIS WILL ALLOW THE PORTION OF PIPE WITHIN THE BASINS TO BE PUMPED DRY BEFORE CUTTING. PUMP WASTEWATER FROM THE INFLUENT SAMPLE PORT TO TREATMENT PLANT #1 AND PUMP ALL FLOW FROM UPSTREAM MANHOLE TO TREATMENT PLANTS 1 AND 2. CONTRACTOR TO PROVIDE A TEMPORARY BYPASS SYSTEM PER SPECS. DURING THIS TIME PIPING MAY BE CUT, INSTALL FLOW METER PIPING, AND DISCHARGE CONNECTIONS CAN BE MADE, EXISTING PIPE PENETRATIONS SHALL BE FILLED AND ALLOWED ADEQUATE TIME TO CURE.

4.BEFORE THE INFLUENT BASIN IS READY TO ACCEPT WASTEWATER THE FOLLOWING CHECKLIST MUST BE COMPLETED:

a.INTERIOR CONCRETE WALLS AND COATING ARE CURED AND READY TO ACCEPT WASTEWATER

b.PIPE PENETRATIONS HAVE BEEN SEALED AND GROUTING HAS CURED.

c.ALL THREE WEIR GATES ARE INSTALLED AND SET TO THE APPROPRIATE LEVELS. WEIR GATES #1 AND #3 SHOULD BE COMPLETELY CLOSED (IN THE HIGH POSITION).

d.DISCHARGE PIPING FROM DISCHARGE BASIN #2 IS COMPLETE.

e.INFLUENT VALVE TO PLANTS ARE OPEN

5.REMOVE UPSTREAM MANHOLE PLUG TO BEGIN ACCEPTING WASTEWATER.

DEMOLITION SCHEDULE:

1. REMOVE & DISPOSE OF EXISTING 8" SLAB OVER ORIGINAL PRE-AERATION BASIN. BASIN & SLAB EXTEND UNDER PLANT BUILDING. CONTRACTOR SHALL SAW CUT CONCRETE SLAB AS CLOSE TO EAST FACE OF BUILDING AS POSSIBLE, LEAVING PORTION OF 8" SLAB THAT EXTENDS UNDER BUILDING.

2. LEAVING EXISTING INTERNAL PIPING INTACT, CONTRACTOR SHALL REMOVE AND DISPOSE OF ALL EXISTING FILL FROM EXISTING BASIN. SUPPORT INTERNAL PIPING AS NECESSARY.

3. PER CONSTRUCTION PHASING SCHEDULE, CONTRACTOR SHALL REMOVE INTERNAL PIPING AFTER NEW FLOW SPLITTER IS OPERATIONAL.

GENERAL NOTES:

CONTRACTOR TO PREPARE A DEMOLITION PLAN WITH SCHEDULE, AND PRESENT TO ENGINEER FOR COORDINATION AND APPROVAL PRIOR TO BEGINNING ANY DEMOLITION ACTIVITIES.

811

Call 811

two business days before you dig

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JGC

DRAWN BY

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AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DEMOLITION PLAN

SHEET

C0.5

PAGE

12

DATE

NOV 2, 2022

SCALE

AS SHOWN

JOB NUMBER

2020-070

OF

96

11-02-2022

PLOT SETTINGS: Adobe PDF1,pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C0.6 STAGING PLANDWG - 11/2/2022 3:21 PM - Jeff Smith

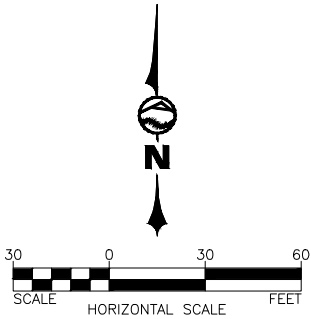


NO.	REVISIONS	BY	DATE
1	NEW STAGING AREA	JGC	5-10-22

- LEGEND
- SF = SILT FENCE PER 3 C0.4
 - = EXISTING GROUND SLOPE
 - = EXISTING DITCH FLOW
 - = VEHICLE PATH

- KEYED NOTES
- 1 = STAGING AREA
 - 2 = EXISTING VEGETATED DITCH

SHEET NOTE:
1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.



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two business days
before you dig

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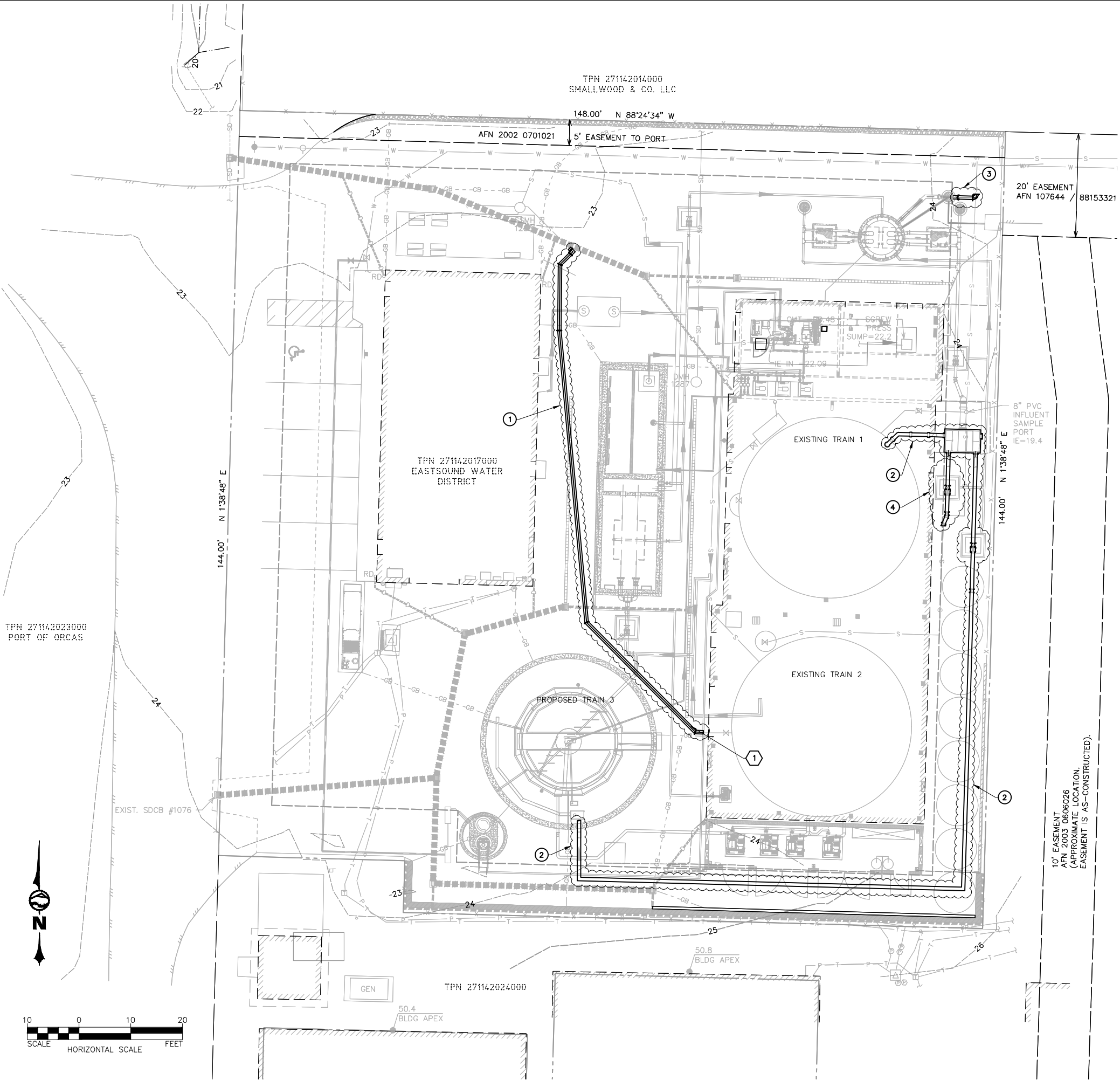
EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY JGC	
SAN JUAN COUNTY		DRAWN BY JGS	
WASHINGTON		CHECKED BY AWL	
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I			
STAGING PLAN			
DATE NOV 2, 2022	SCALE AS SHOWN	JOB NUMBER 2020-070	
SHEET C0.6	PAGE 13	OF 96	

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W:\2020-2070 EASTSOUND WWTP DESIGN\DWG\2020-070 C0.7 INTERIM PLAN OF OPERATION.DWG - 11/2/2022 3:21 PM - Jeff Smith



- KEYED NOTES**
- 1 = PROVIDE TEMPORARY 8" PIPING FROM THE EXISTING CHLORINE DISINFECTION OUTLET (PIPE DIAMETER = 10") TO THE EXISTING 8" GRAVITY SEWER (JUST UPSTREAM OF SSMH 1288). TEMPORARY PIPE IS TO BE FLEXIBLE HDPE PIPE WITH RESTRAINED BAUER END CONNECTIONS OR APPROVED EQUAL.
 - ENSURE POSITIVE DRAINAGE FOR TEMPORARY 8" PIPE. MIN SLOPE = 0.04%.
 - POT HOLE EXISTING PIPE AND PROVIDE PLANNED ALIGNMENT INFORMATION FOR REVIEW AND APPROVAL, PRIOR TO CONSTRUCTION.
 - ENSURE WATER TIGHT RESTRAINED CONNECTIONS ARE PROVIDED AT UPSTREAM AND DOWNSTREAM CONNECTIONS.

CONSTRUCTION SEQUENCE:
NOTE: ALL CONSTRUCTION PHASING STEPS SHALL BE COORDINATED WITH THE OWNER AND ENGINEER.

- 1 = PRIOR TO CONSTRUCTION, DIVERT TREATED EFFLUENT AROUND THE PROPOSED WORK AREA THROUGH A TEMPORARY 8" PIPE. SEE KEYED NOTE #1
- 2 = INSTALL NEW SPLITTER BOX AND PIPING TO TRAIN #1 AND TRAIN #3 WITHOUT DISTURBING EXISTING PIPING AND PLANT OPERATIONS.
- 3 = WHEN STAGE 1 FACILITIES ARE COMPLETE, TESTED, ACCEPTED AND READY TO BE PLACED ON-LINE, CONNECT 8" INFLUENT TO NEW INFLUENT PUMP STATION AND DIRECT FLOW TO TRAIN #1.
- 4 = DEWATER, ABANDON, AND REMOVE EXISTING INFLUENT PIPING CONFLICTS SO THAT TRAIN #2 PIPING AND FLOW METER VAULT CAN BE INSTALLED AND TRAIN #2 PLACED IN SERVICE. TRAIN #2 DOWNTIME IS TO BE LIMITED TO 2-DAYS MAX, AND TO BE SCHEDULED DURING DRY WEATHER CONDITIONS. COORDINATE WITH OWNER.
- 5 = AFTER TREATED EFFLUENT FLOW HAS BEEN DIVERTED TO NEW UV DISINFECTION SYSTEM, PLUG & ABANDON THE TEMPORARY 8" PIPE (IDENTIFIED AS KEYED NOTE #1).
- 6 = PLACE STAGE 2 FACILITIES ON-LINE.
- 7 = PLACE STAGE 3 FACILITIES ON-LINE.

STAGES:

STAGE 1: INFLUENT PUMP STATION, INFLUENT SAMPLE STATION, INFLUENT FLOW SPLITTER, AEROBIC DIGESTER AND BLOWER, SLUDGE PUMP SYSTEM, UV DISINFECTION SYSTEM, BLOWER BUILDING.

STAGE 2: SLUDGE DEWATERING SYSTEM, 2W WATER SYSTEM.

STAGE 3: TRAIN #3.

SHEET NOTES
1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.

NO.	REVISIONS	BY	DATE
5	ADD INTERIM PLAN OF OP.	JGC	5-10-22

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I	
WWTP INTERIM PLAN OF OPERATION	
DATE NOV 2, 2022	SCALE AS SHOWN
SHEET C0.7	JOB NUMBER 2020-070
PAGE 14	OF 96



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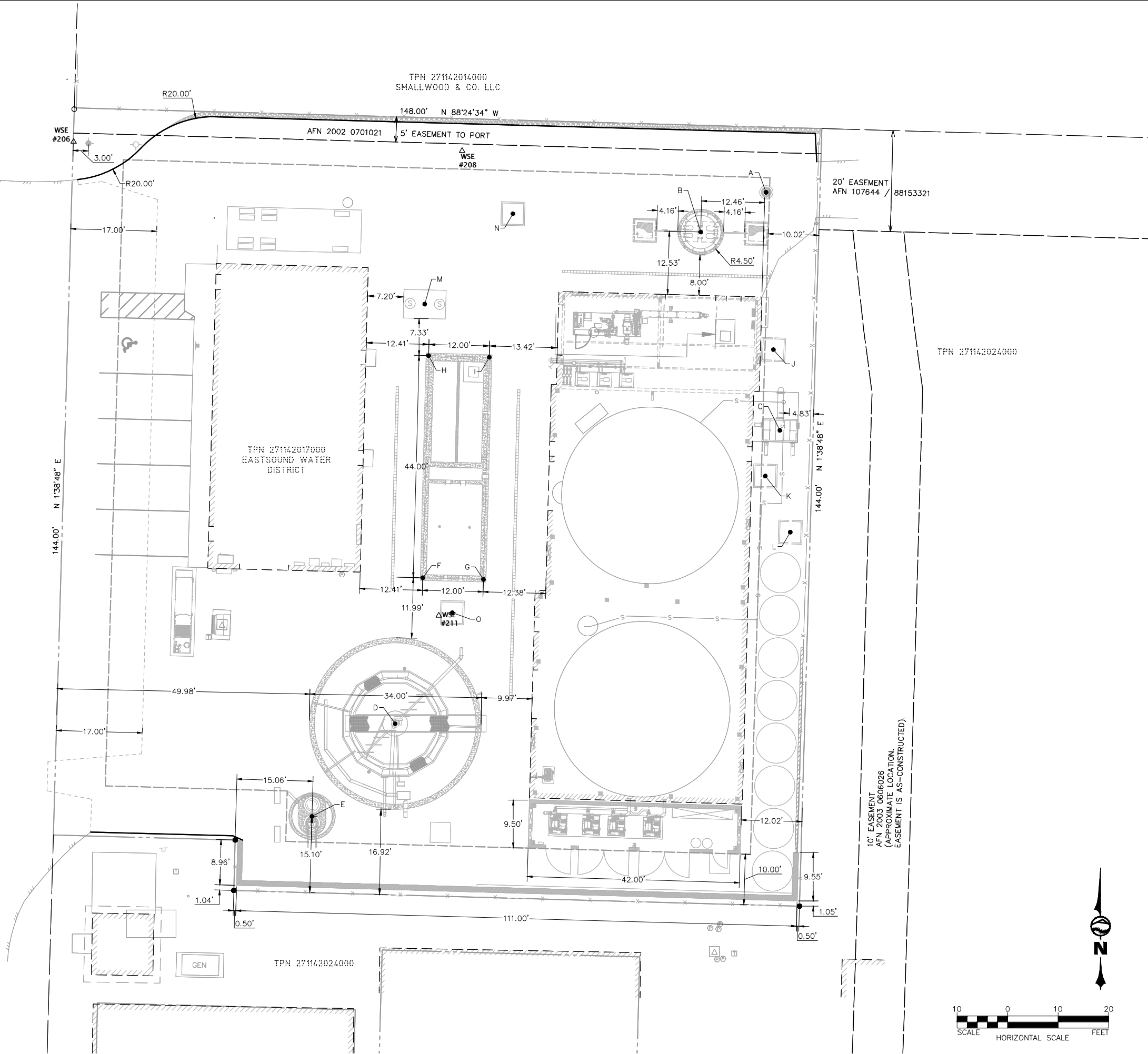
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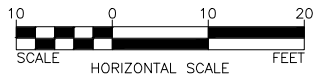
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TPN 271142023000
PORT OF ORCAS

WSE
#205



10' EASEMENT
AFN 2003 0606026
(APPROXIMATE LOCATION,
EASEMENT IS AS-CONSTRUCTED).



NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

COORDINATES:

CONTROL:

WSE 205: N 629,477.19 E 1,139,257.92
WSE 206: N 629,622.12 E 1,139,287.90
WSE 208: N 629,620.32 E 1,139,364.78
WSE 211: N 629,528.31 E 1,139,360.16

NEW STRUCTURES:

A: INFL SAMPLE SUMP
N 629,612.08 E 1,139,424.97
B: INFL PS WET WELL
N 629,604.23 E 1,139,412.01
C: INFL FLOW SPLITTER
N 629,564.98 E 1,139,427.67
D: TRAIN #3 CENTER
N 629,506.93 E 1,139,351.54
E: 2W PS WET WELL
N 629,488.55 E 1,139,335.08
F: SW CORNER
N 629,535.78 E 1,139,357.01
G: SE CORNER
N 629,535.46 E 1,139,369.01
H: NW CORNER
N 629,579.76 E 1,139,358.17
I: NE CORNER
N 629,579.45 E 1,139,370.17
J: INFL FM CENTER
N 629,580.96 E 1,139,426.40
K: TRAIN #2 FM CENTER
N 629,555.94 E 1,139,424.79
L: TRAIN #3 FM CENTER
N 629,544.82 E 1,139,429.74
M: SEPTIC TANK CENTER
N 629,589.95 E 1,139,357.39
N: PLANT DRAIN FM CENTER
N 629,607.85 E 1,139,374.86
O: EFFLUENT FM CENTER
N 629,528.89 E 1,139,362.85

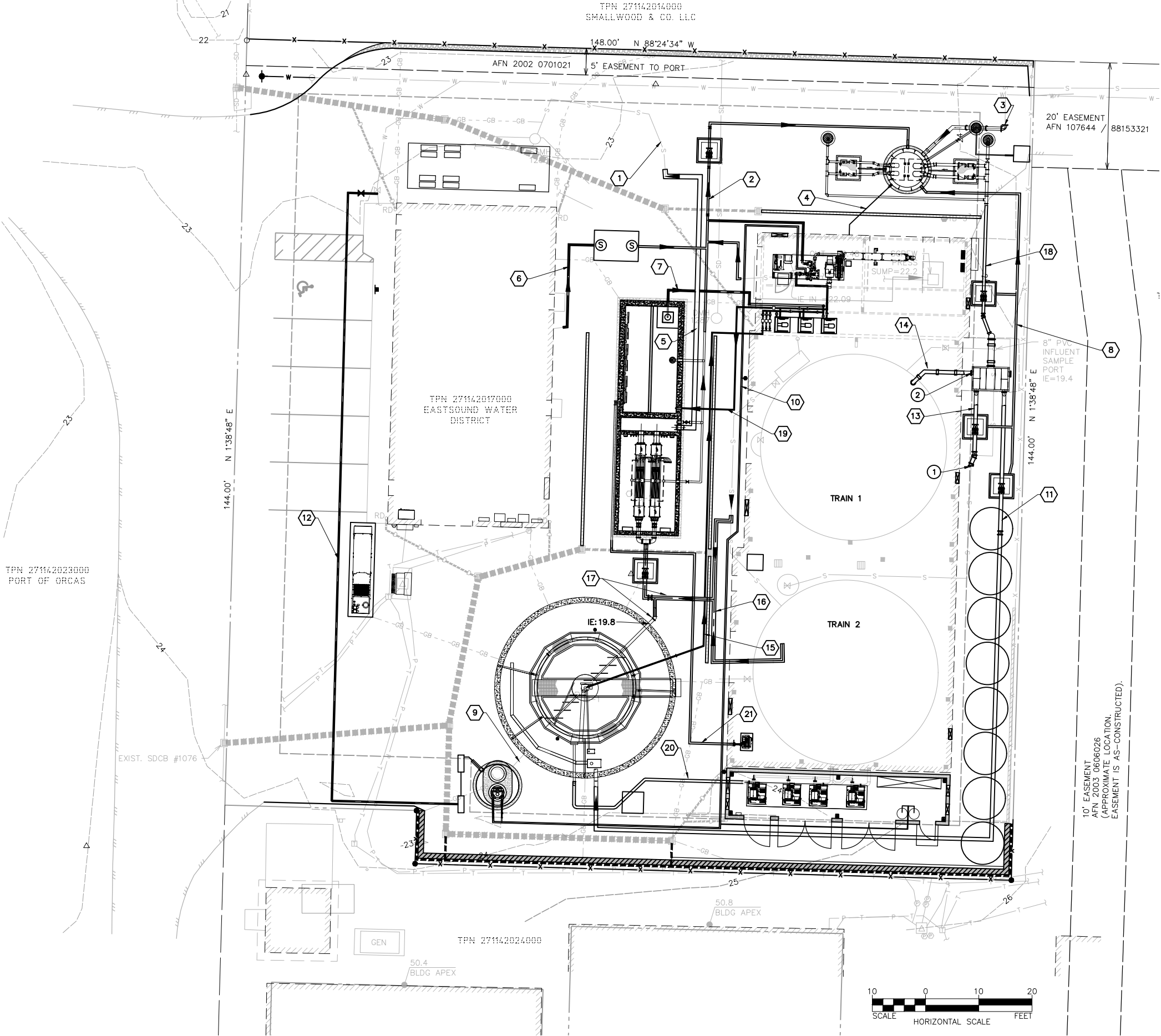


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C1.2		NOV 2, 2022	WASHINGTON		JCC
		SCALE	SAN JUAN COUNTY		DRAWN BY
		AS SHOWN	WASTEWATER TREATMENT PLANT UPGRADE – PHASE I		SNR
		JOB NUMBER	SITE DIMENSIONAL PLAN		CHECKED BY
		2020–070			AWL
16 OF 96					

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TPN 271142023000
PORT OF ORCAS



PIPE IDENTIFICATION LEGEND	
1	8" EXISTING EFFLUENT (DI GRAV.)
2	4" UV DISINFECTION DRAIN AND DIGESTER SUPERNATANT (DI GRAV.) SLOPE = 2% MIN.
3	8" INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
4	3" INFLUENT PUMP STATION AIR VENT (SS) OUTLET 8" ABOVE FINISH GRADE WITH GOOSENECK POINTING DOWN. ATTACH PIPE SECURELY TO EXTERIOR BUILDING WALL SLOPE = POSITIVE DRAINAGE THROUGHOUT
5	10" UV DISINFECTION EFFLUENT (DI GRAV.) SLOPE = 0.28% MIN.
6	4" BUILDING SEPTIC (SCH. 80 PVC GRAV.) CLOPE = 2% MIN.
7	2" DIGESTED SLUDGE (SCH. 80 PVC FM) SLOPE = POSITIVE THROUGHOUT
8	4" VAULT DRAIN (SCH. 80 PVC GRAV.) SLOPE = 2.0%
9	1" 2W WATER SYSTEM AIR VENT (SS) OUTLET 3' ABOVE GRADE WITH GOOSENECK POINTING DOWN SLOPE = POSITIVE DRAINAGE THROUGHOUT
10	2" 2W WATER LINE (SCH. 80 PVC FM)
11	8" TRAIN 3 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
12	2" WATER PIPE (SCH. 80 PVC FM)
13	8" TRAIN 2 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
14	8" TRAIN 1 INFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
15	2" TRAIN 3 WAS (SCH. 80 PVC GRAV.) SLOPE = POSITIVE DRAINAGE THROUGHOUT
16	8" TRAIN 2 EFFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
17	8" TRAIN 3 EFFLUENT (DI GRAV.) SLOPE = 0.4% MIN.
18	6" INFLUENT TO FLOW SPLITTER (DI FM) SLOPE = POSITIVE THROUGHOUT
19	2" DIGESTER WAS INFLUENT (SCH. 80 PVC GRAV.)
20	6" CELL 3 AIR (SS)
21	4" DIGESTER AIR (SS)

- KEYED NOTES**
- ① = CUT EXISTING 8" PIPE AND CONNECT NEW 8" INFLUENT PIPE TO TRAIN 2
- ② = EXTEND NEW 8" PIPE TO TRAIN 1. ABANDON AND PLUG EXISTING TRAIN 1 INFLUENT PIPE

- SHEET NOTES**
- 1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.
- 2) SEE SHEETS C2.1-C2.4 FOR STORM GRADING, DRAINAGE, AND PIPING.



BID SET

NO.	REVISIONS	BY	DATE
2	ADD INF. SAMPLE SUMP	JGC	5/10/22



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DRAWN BY	SNR
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

SITE PIPING PLAN WITH PIPING ID

SHEET	C1.3
DATE	NOV 2, 2022
SCALE	AS SHOWN
JOB NUMBER	2020-070
PAGE	17 OF 96

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EASTSOUND SEWER AND WATER DISTRICT	
SAN JUAN COUNTY	WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I	
SITE PIPING PLAN WITH BURIED VALVE ID	

SHEET	C1.4	DATE	NOV 2, 2022
		SCALE	AS SHOWN
PAGE	18 OF 96	JOB NUMBER	2020-070

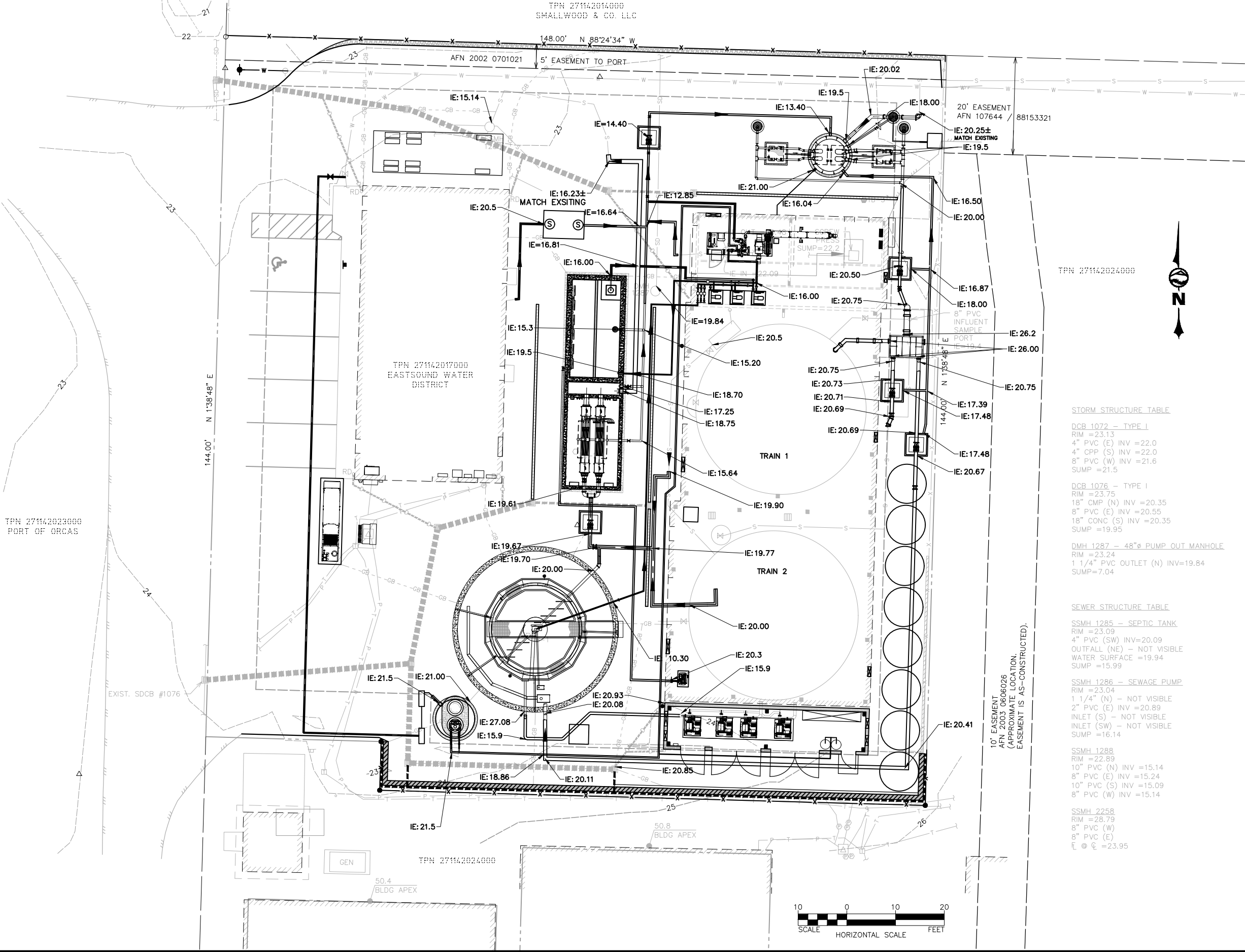


- 1) SEE SHEET C0.1 FOR EXISTING CONDITIONS.
- 2) SEE SHEETS C2.1-C2.4 FOR STORM GRADING, DRAINAGE, AND PIPING.



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TPN 271142023000
PORT OF ORCAS



STORM STRUCTURE TABLE

DCB 1072 - TYPE I
RIM =23.13
4" PVC (E) INV =22.0
4" CPP (S) INV =22.0
8" PVC (W) INV =21.6
SUMP =21.5

DCB 1076 - TYPE I
RIM =23.75
18" CMP (N) INV =20.35
8" PVC (E) INV =20.55
18" CONC (S) INV =20.35
SUMP =19.95

DMH 1287 - 48" PUMP OUT MANHOLE
RIM =23.24
1 1/4" PVC OUTLET (N) INV=19.84
SUMP=7.04

SEWER STRUCTURE TABLE

SSMH 1285 - SEPTIC TANK
RIM =23.09
4" PVC (SW) INV=20.09
OUTFALL (NE) - NOT VISIBLE
WATER SURFACE =19.94
SUMP =15.99

SSMH 1286 - SEWAGE PUMP
RIM =23.04
1 1/4" (N) - NOT VISIBLE
2" PVC (E) INV =20.89
INLET (S) - NOT VISIBLE
INLET (SW) - NOT VISIBLE
SUMP =16.14

SSMH 1288
RIM =22.89
10" PVC (N) INV =15.14
8" PVC (E) INV =15.24
10" PVC (S) INV =15.09
8" PVC (W) INV =15.14

SSMH 2258
RIM =28.79
8" PVC (W)
8" PVC (E)
E @ E =23.95

10' EASEMENT
AFN 2003 0606026
(APPROXIMATE LOCATION,
EASEMENT IS AS-CONSTRUCTED).

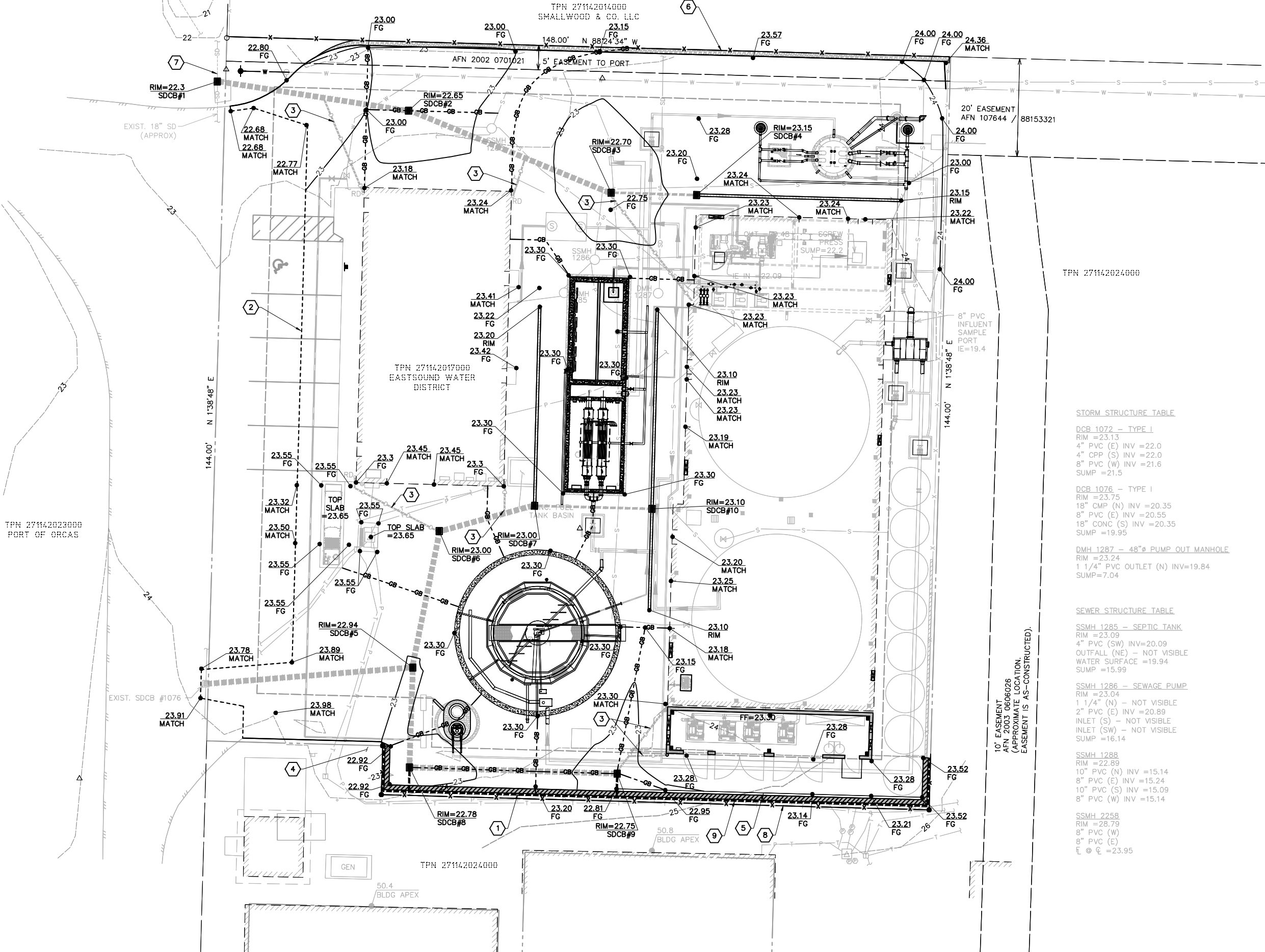


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C1.5		NOV 2, 2022		JCC					
PAGE		SCALE		DRAWN BY		SNR		CHECKED BY	
		AS SHOWN		SAN JUAN COUNTY		WASHINGTON		AWL	
		JOB NUMBER		WASTEWATER TREATMENT PLANT UPGRADE – PHASE I					
19 OF 96		2020–070		SITE PIPING PLAN WITH PIPE ELEVATIONS					

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TPN 271142023000
PORT OF ORCAS



STORM STRUCTURE TABLE

DCB 1072 - TYPE I
RIM = 23.13
4" PVC (E) INV = 22.0
4" CPP (S) INV = 22.0
8" PVC (W) INV = 21.6
SUMP = 21.5

DCB 1076 - TYPE I
RIM = 23.75
18" CMP (N) INV = 20.35
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RIM = 23.09
4" PVC (SW) INV = 20.09
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SSMH 1286 - SEWAGE PUMP
RIM = 23.04
1 1/4" (N) - NOT VISIBLE
2" PVC (E) INV = 20.89
INLET (S) - NOT VISIBLE
INLET (SW) - NOT VISIBLE
SUMP = 16.14

SSMH 1288
RIM = 22.89
10" PVC (N) INV = 15.14
8" PVC (E) INV = 15.24
10" PVC (S) INV = 15.09
8" PVC (W) INV = 15.14

SSMH 2258
RIM = 28.79
8" PVC (W)
8" PVC (E)
E @ E = 23.95

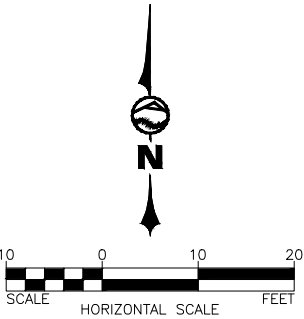
NO.	REVISIONS	BY	DATE
4	ADD TRENCH DRAIN	JGC	5-10-22

KEYED NOTES

- 1 = KEYSTONE BLOCK RETAINING WALL. SEE PLAN, PROFILE, AND DETAILS PER C2.5 & C7.8
- 2 = SAWCUT LINE PER DEMOLITION PLAN C0.5
- 3 = 4" DIA ROOF DRAIN
- 4 = GRADE AREA FROM FINISHED GRADES TO DAYLIGHT AT 2H:1V
- 5 = ASPHALT WEDGE CURB PER C7.7
- 6 = 149 LF OF 6' CHAIN LINK FENCE. CONNECT TO EXISTING.
- 7 = PRESUMED LOCATION OF EXISTING STORM SEWER. CATCH BASIN LOCATION TO BE ADJUSTED BASED ON POTHOLES RESULTS
- 8 = 134 LF OF 6' CHAIN LINK FENCE. CONNECT TO EXISTING
- 9 = FOOTING DRAIN

NOTES:

- 1) MATCH ELEVATION AT BUILDING DOORS IS 0.02' BELOW FINISHED FLOOR.
- 2) ALL CATCH BASINS TO HAVE HERRINGBONE GRATES.
- 3) ALL HATCHES AND COVERS FOR NON-STORMWATER ITEM 5 TO BE 0.02' MIN ABOVE SURROUNDING ASPHALT GRADES. IN CASES WHERE HATCHES & COVERS ARE GREATER THAN 0.02' ABOVE SURROUNDING ASPHALT, SLOPE ASPHALT AT 1H:1V, AS NEEDED TO ACHIEVE 0.02' DISTANCE.
- 4) SEE SHEET C0.1 FOR EXISTING CONDITIONS.



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DATE
NOV 2, 2022

SCALE
AS SHOWN

SHEET
C2.1

SAN JUAN COUNTY

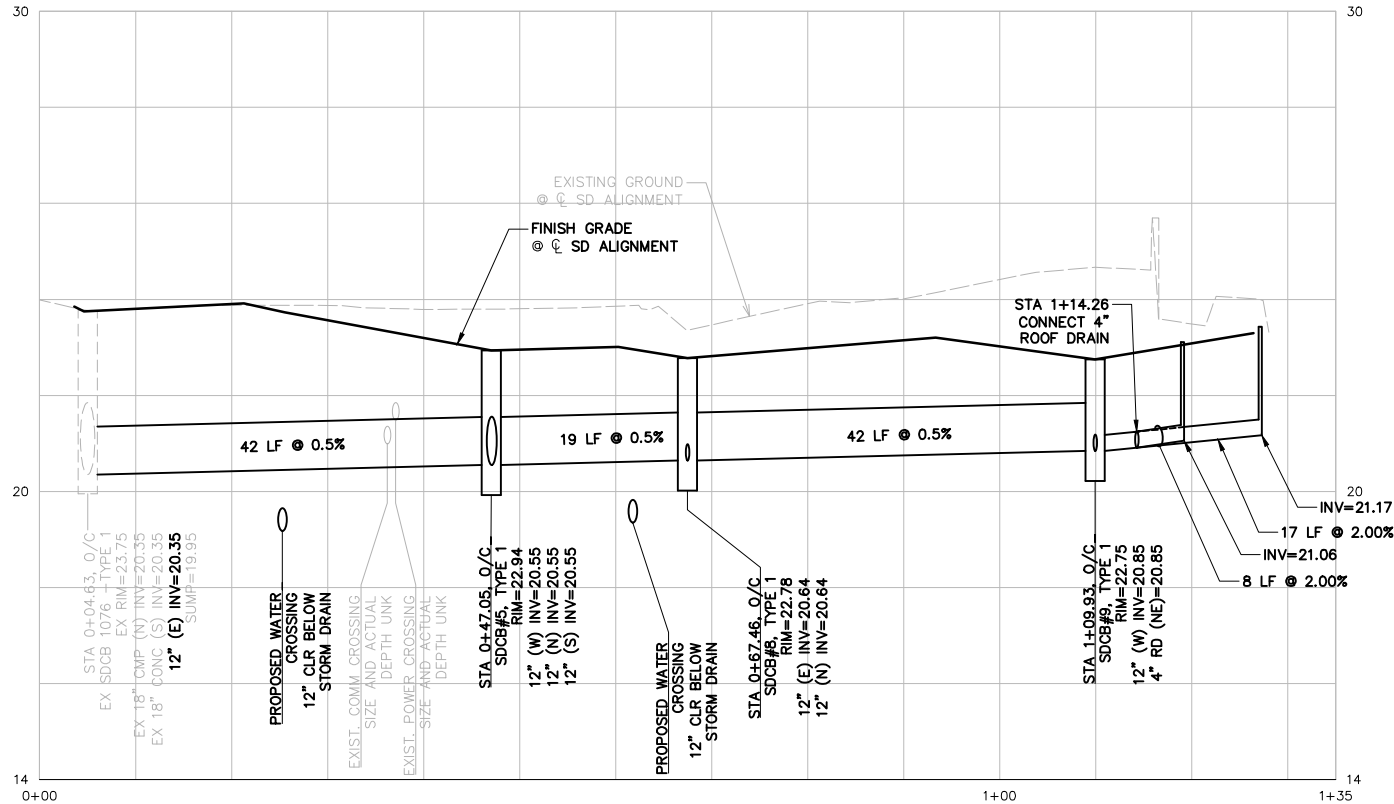
JOB NUMBER
2020-070

PAGE
20 OF 96

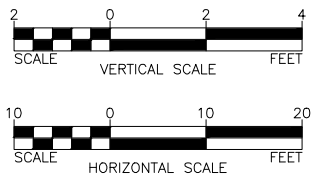
GRADING & STORM DRAINAGE PLAN

11-02-2022

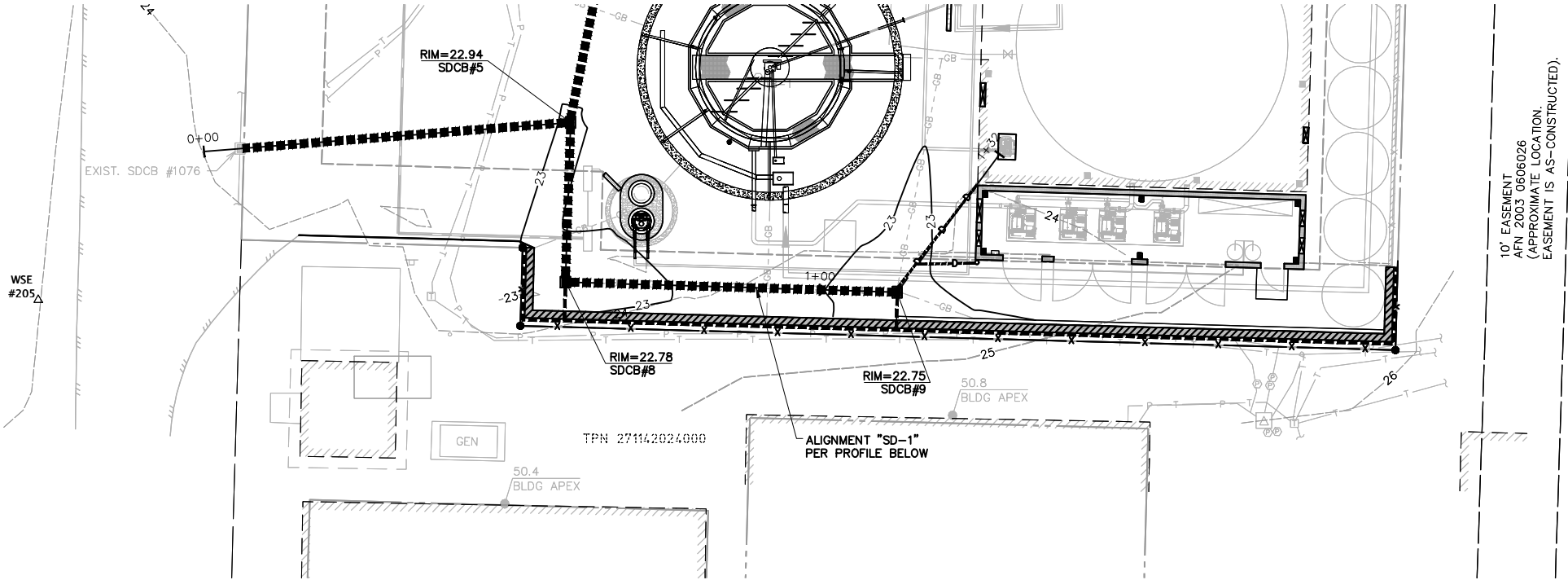
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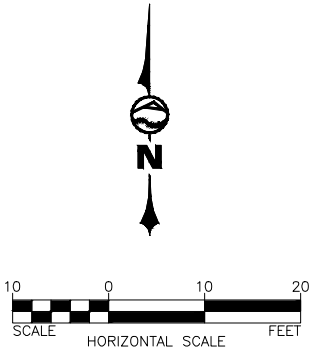
ALIGNMENT "SD-1" - PROFILE VIEW



BID SET



ALIGNMENT "SD-1" - PLAN VIEW



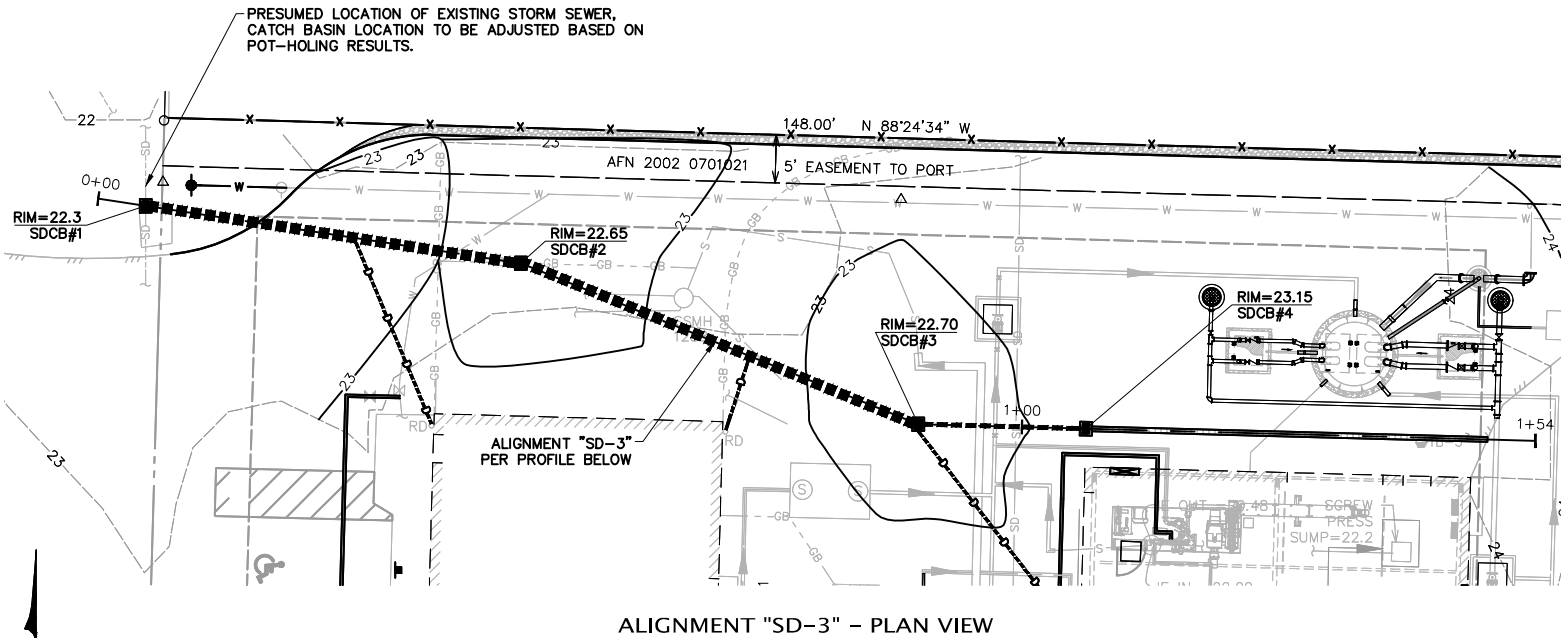
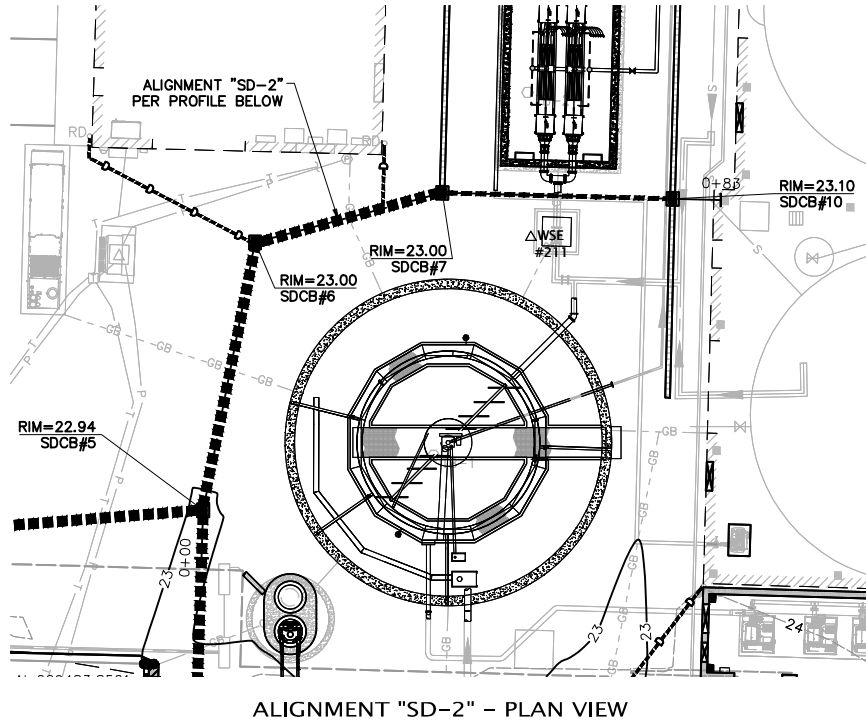
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SHEET C2.2	DATE NOV 2, 2022	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY JCC	G. CHIEF ENGINEER STATE OF WASHINGTON REG. NO. 35585 PROFESSIONAL	WILSON ENGINEERING WILSONENGINEERING.COM
	SCALE AS SHOWN	SAN JUAN COUNTY WASHINGTON WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		DRAWN BY JGS		
PAGE 21 OF 96	JOB NUMBER 2020-070	STORM DRAINAGE PLAN & PROFILE		CHECKED BY AWL		

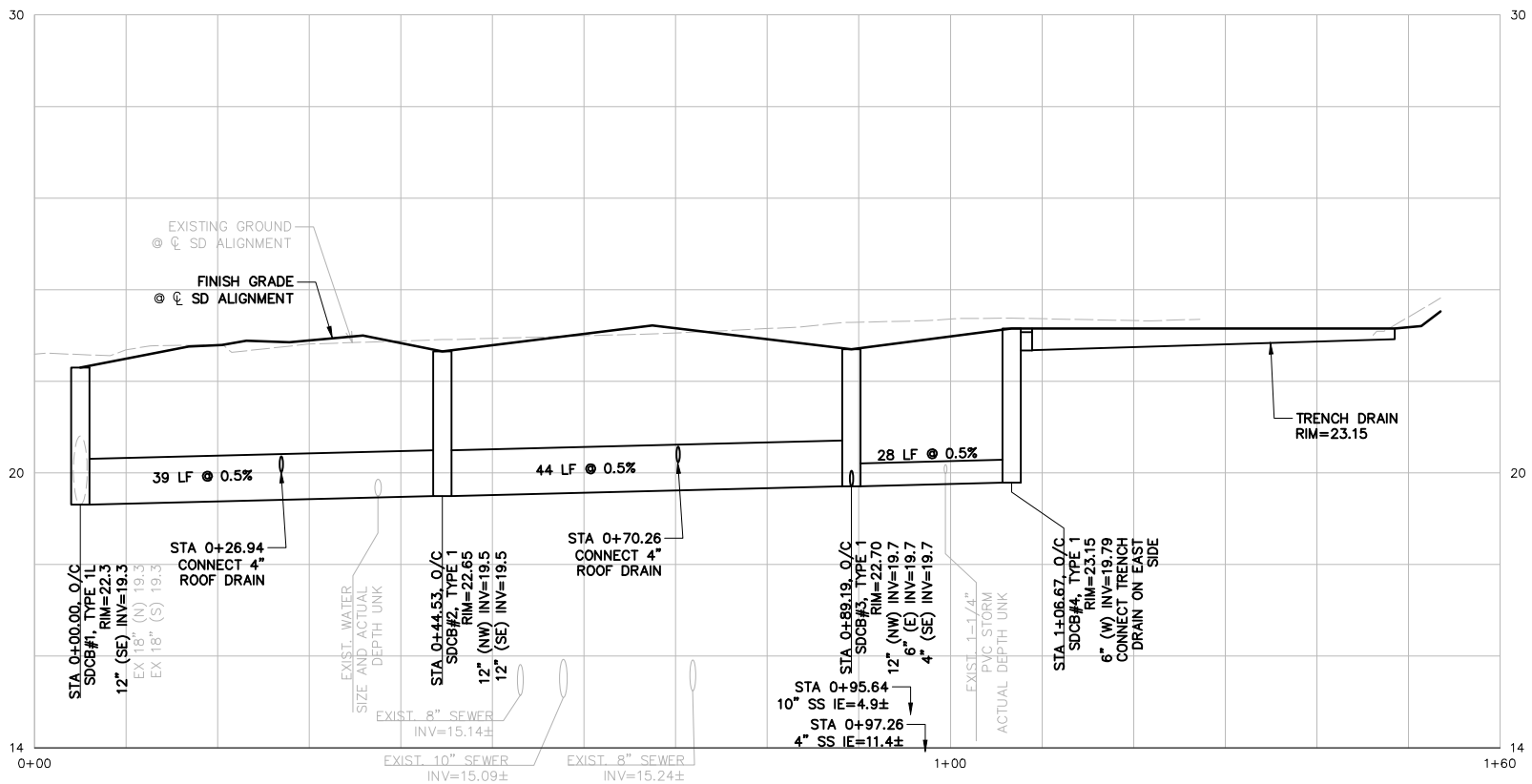
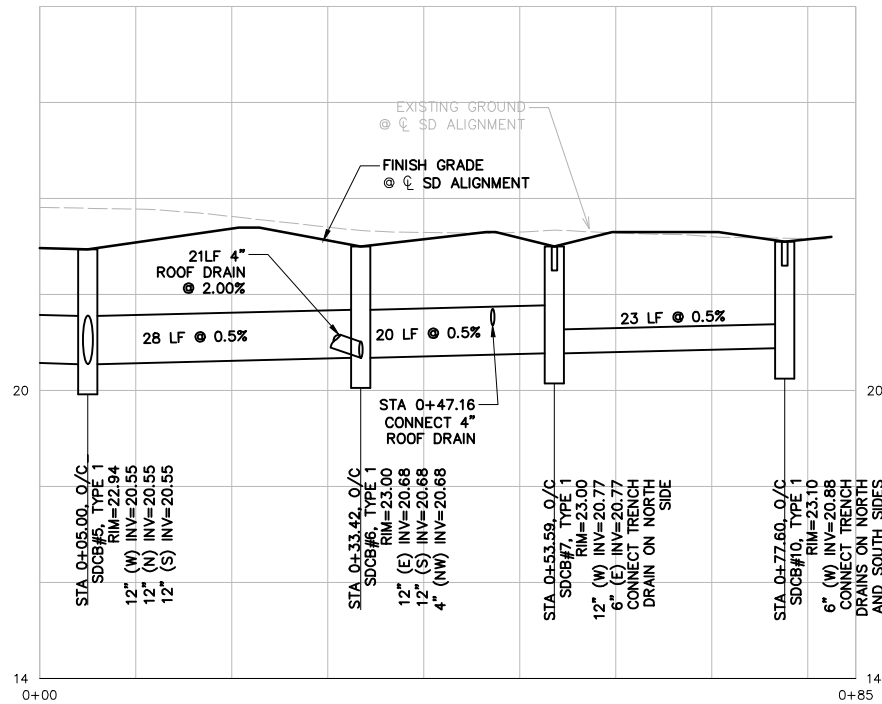
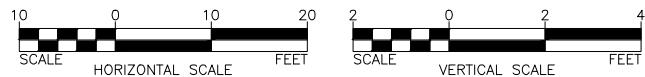
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NO.	REVISIONS	BY	DATE
4	ADD TRENCH DRAIN	JGC	5-10-22



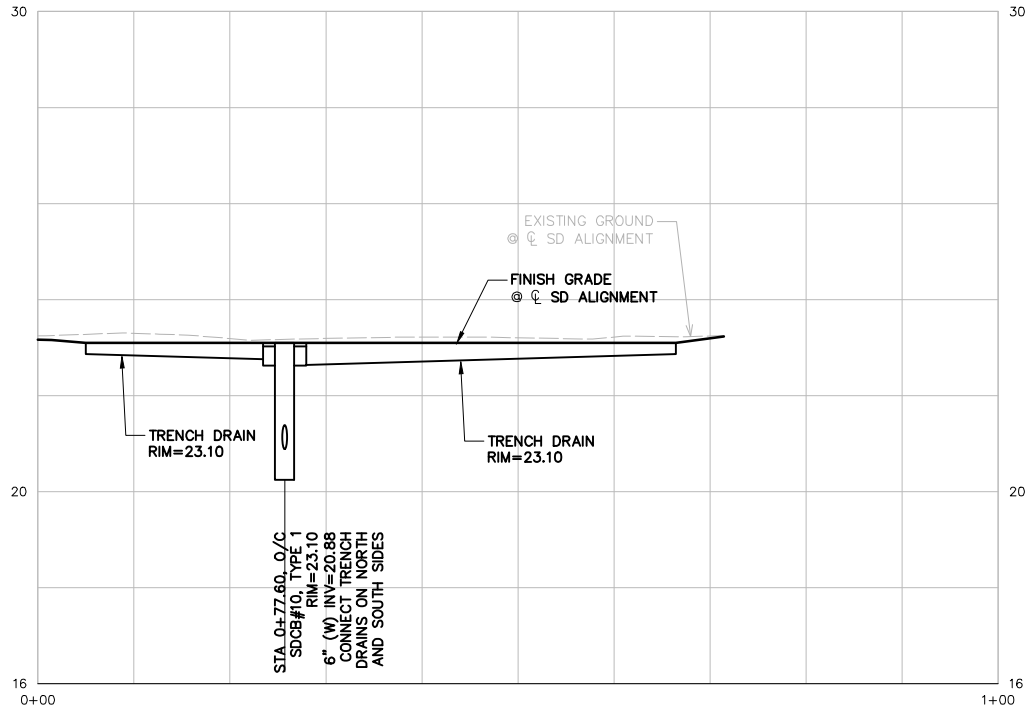
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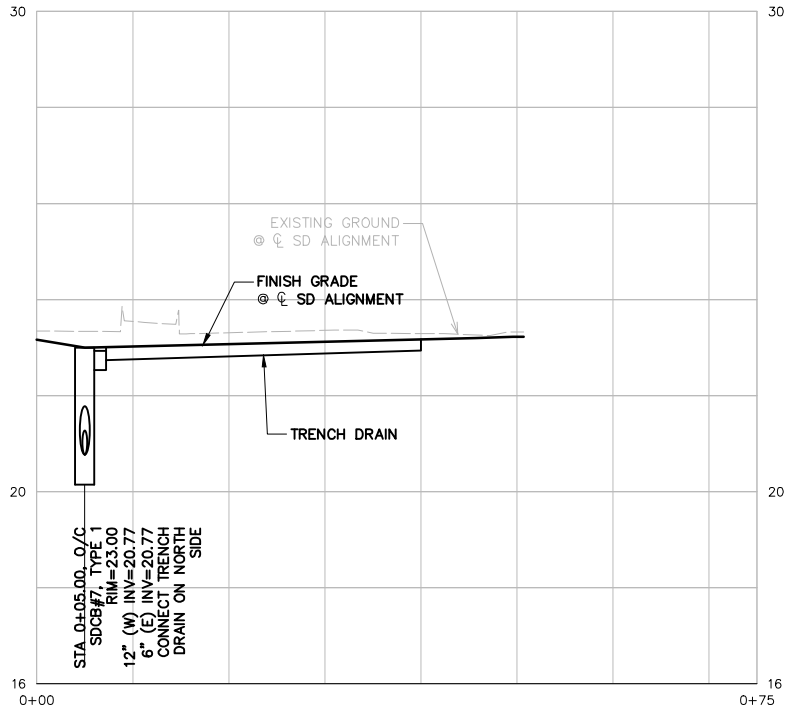
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	SCALE AS SHOWN	SAN JUAN COUNTY WASHINGTON		DRAWN BY JGS	
	PAGE 22 OF 96	WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		CHECKED BY AWL	
		STORM DRAINAGE PLAN & PROFILES		WILSONENGINEERING.COM	

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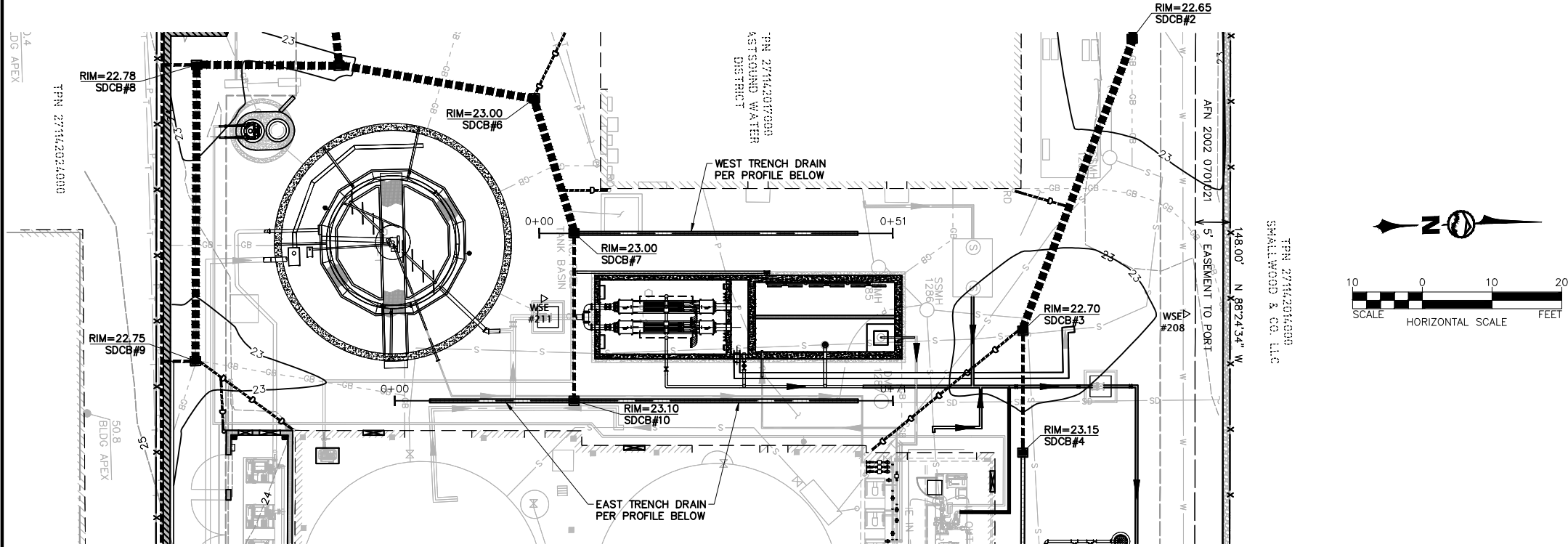
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EAST TRENCH DRAIN – PROFILE VIEW



WEST TRENCH DRAIN – PROFILE VIEW



WEST & EAST TRENCH DRAINS – PLAN VIEW

NO.	REVISIONS	BY	DATE
4	ADD TRENCH DRAIN	JGC	5-10-22



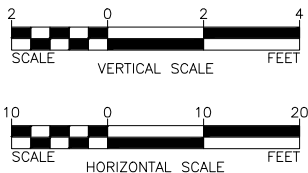
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CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

STORM DRAINAGE PLAN & PROFILES

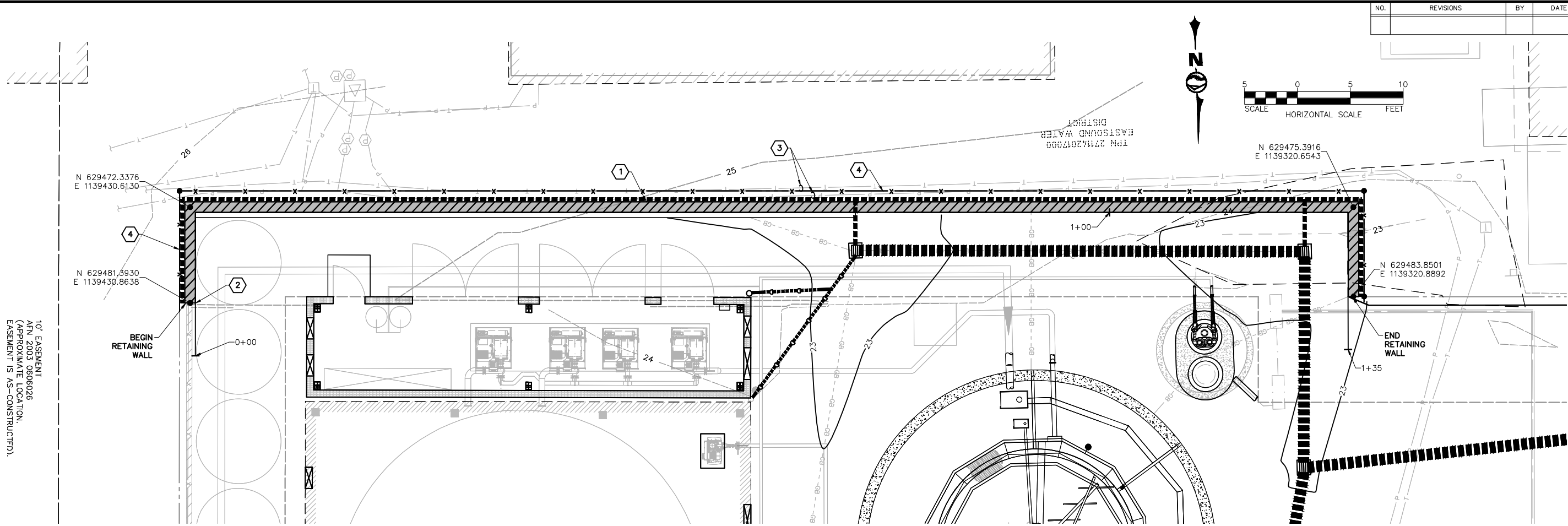
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DATE	NOV 2, 2022
SCALE	AS SHOWN
JOB NUMBER	2020-070
PAGE	23
OF	96



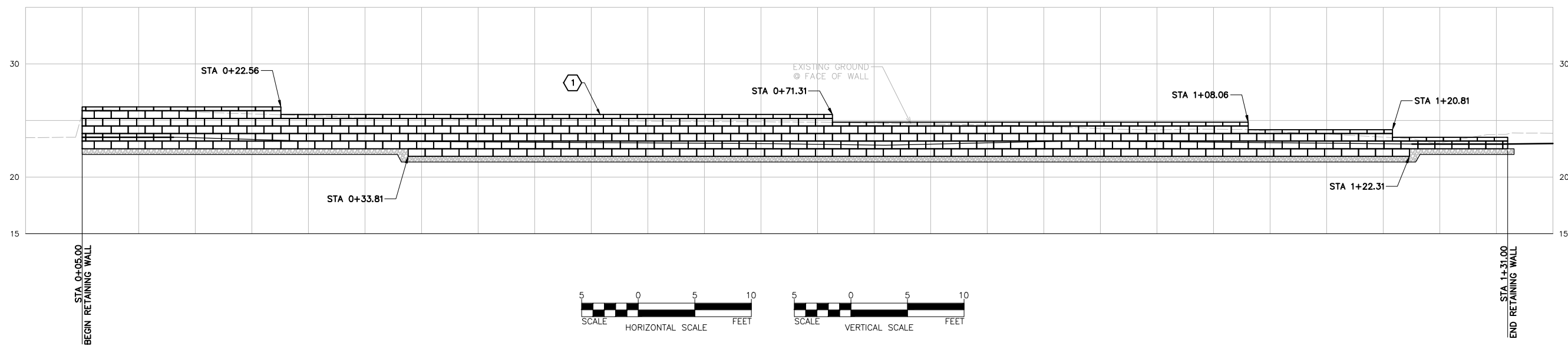
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- KEYED NOTES**
- 1 = BLOCK RETAINING WALL PER $\frac{1}{C7.8}$
 - 2 = PROVIDE WATERPROOF EXPANSIVE FOAM TO FILL GAP BETWEEN PROPOSED AND EXISTING WALLS
 - 3 = RELOCATED POWER & COMMUNICATIONS AS NEEDED PER DEMOLITION PLAN
 - 4 = 126 LF OF 6' CHAIN LINK FENCE. CONNECT TO EXISTING FENCE PER SPECIFICATIONS
- NOTE: COORDINATES ARE AT THE CENTER OF BLOCK ADJACENT TO FINISHED ASPHALT GRADE. SEE $\frac{5}{C7.7}$

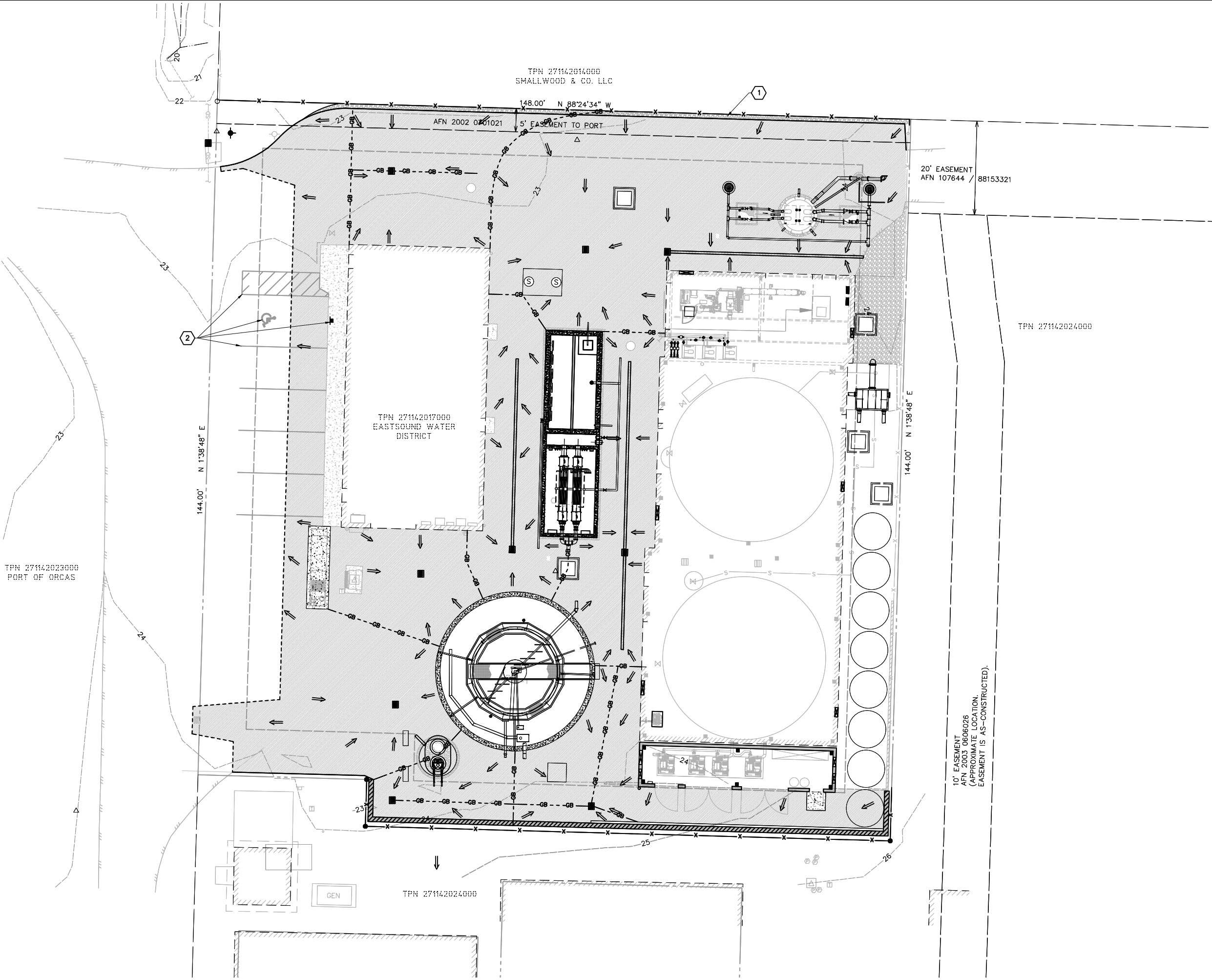


811 Call 811
two business days
before you dig

BID SET

SHEET C2.5	DATE NOV 2, 2022	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY JCC DRAWN BY JCS CHECKED BY AWL		WILSON ENGINEERING WILSONENGINEERING.COM
	SCALE AS SHOWN	SAN JUAN COUNTY WASHINGTON WASTEWATER TREATMENT PLANT UPGRADE - PHASE I				
PAGE 24 OF 96	JOB NUMBER 2020-070			11-02-2022		

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
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NO.	REVISIONS	BY	DATE
4	ADD TRENCH DRAIN	JGC	5-10-22

PAVING PLAN LEGEND

- = PROPOSED ASPHALT PAVING PER $\frac{3}{C7.3}$
- = PROPOSED CONCRETE PER:
 - WALKWAYS= $\frac{1}{C7.2}$
 - GENERATOR PAD= $\frac{4}{S2.4}$
- = PROPOSED GRAVEL PER $\frac{5}{C7.3}$
- = EXISTING CONCRETE TO REMAIN
- = FLOW ARROWS

KEYED NOTES

- 1 = 1' WIDE GRAVEL SHOULDER BETWEEN PROPERTY LINE & ASPHALT
- 2 = PARKING STALL STRIPING, HANDICAP SYMBOL, AND DISABLED PARKING SIGNAGE PER DETAIL $\frac{4}{C7.1}$

10
0
10
20
SCALE HORIZONTAL SCALE FEET

811 Call 811
two business days
before you dig

BID SET

SHEET		DATE	SCALE		JOB NUMBER	
C2.6		NOV 2, 2022	AS SHOWN		2020-070	

DESIGNED BY		DRAWN BY		CHECKED BY	
JGC		JGS		AWL	

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

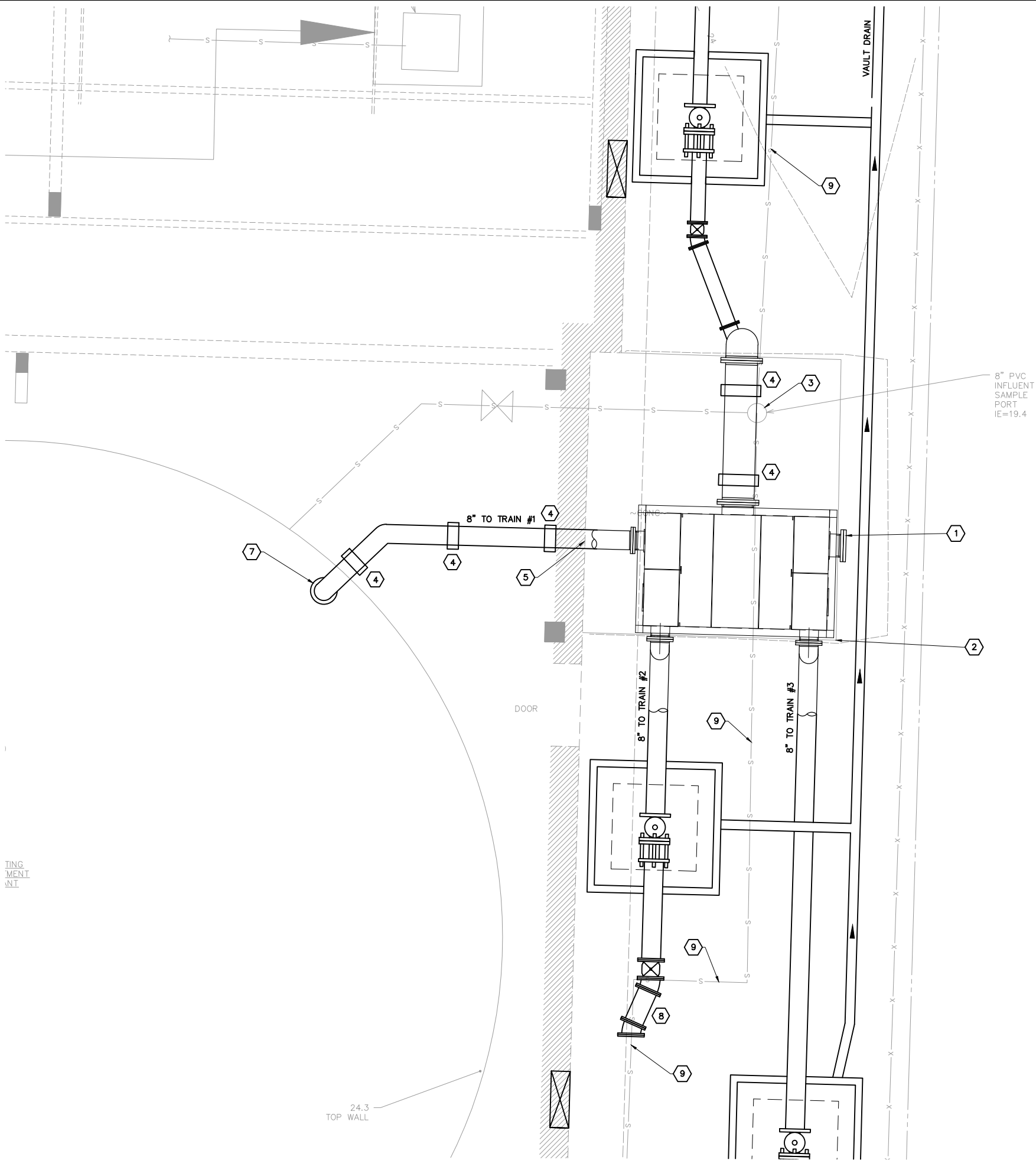
PAVING PLAN

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1 INFLUENT FLOW SPLITTER SITE PLAN



CONSTRUCTION SEQUENCE:

INSTALL NEW SPLITTER BOX AND PIPING TO TRAIN #1 AND TRAIN #3 WITHOUT DISTURBING EXISTING PIPING AND PLANT OPERATIONS. WHEN NEW FACILITIES ARE COMPLETE, TESTED, AND READY TO BE PLACED ON-LINE, CONNECT 8" INFLUENT TO NEW INFLUENT PUMP STATION AND DIRECT FLOW TO TRAINS #1 AND #3. THEN DEWATER, ABANDON, AND REMOVE EXISTING INFLUENT PIPING CONFLICTS SO THAT TRAIN #2 PIPING AND FLOW METER VAULT CAN BE INSTALLED AND TRAIN #2 PLACED IN SERVICE. TRAIN #2 DOWNTIME IS TO BE LIMITED TO 5-DAYS MAX (MONDAY-FRIDAY).

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

KEYED NOTES

- 1 = CONNECTION FOR FUTURE TRAIN #4. SEAL WITH 8" BLIND FLANGE
- 2 = PROPOSED CONCRETE SUPPORT FOR SPLITTER BOX
- 3 = EXISTING SAMPLING PORT
- 4 = PIPE SUPPORT PER $\frac{5}{C7.2}$
- 5 = CAREFULLY CUT HOLE THROUGH BUILDING FOR NEW 8" TRAIN #1 INFLUENT PIPE. SEAL WITH FLASHING
- 6 = NOT USED
- 7 = 8" TRAIN #1 INFLUENT, INSTALL ELBOW AND DIRECT FLOW INTO AERATION BASIN ADJACENT TO EXISTING INFLUENT PIPE
- 8 = CONFIRM ALIGNMENT, SIZE, MATERIAL, AND DEPTH OF EXISTING PIPE TO TRAIN #2 INFLUENT. PROVIDE BENDS & FITTINGS AS NEEDED SO THAT NEW 8" DI PIPING CAN BE CONNECTED TO EXISTING PIPE, AND INFLUENT FLOW IS DIRECTED TO TRAIN #2. ENSURE POSITIVE DRAINAGE THROUGHOUT, FROM SPLITTER BOX TO NEW CONNECTION
- 9 = POT-HOLE EXISTING 8" PIPING AND CONFIRM ACTUAL ALIGNMENT AND DEPTH PRIOR TO CONSTRUCTION. ADJUST FINAL LAYOUT, AS NEEDED, TO AVOID CONFLICTS



BID SET

C3.1	DATE	NOV 2, 2022	EASTSOUND SEWER AND WATER DISTRICT		DESIGNED BY	JCC
	SCALE	AS SHOWN	WASHINGTON		DRAWN BY	JCS
	JOB NUMBER	2020-070	WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		CHECKED BY	AWL
PAGE		26	OF	96	INFLUENT FLOW SPLITTER SITE PLAN	



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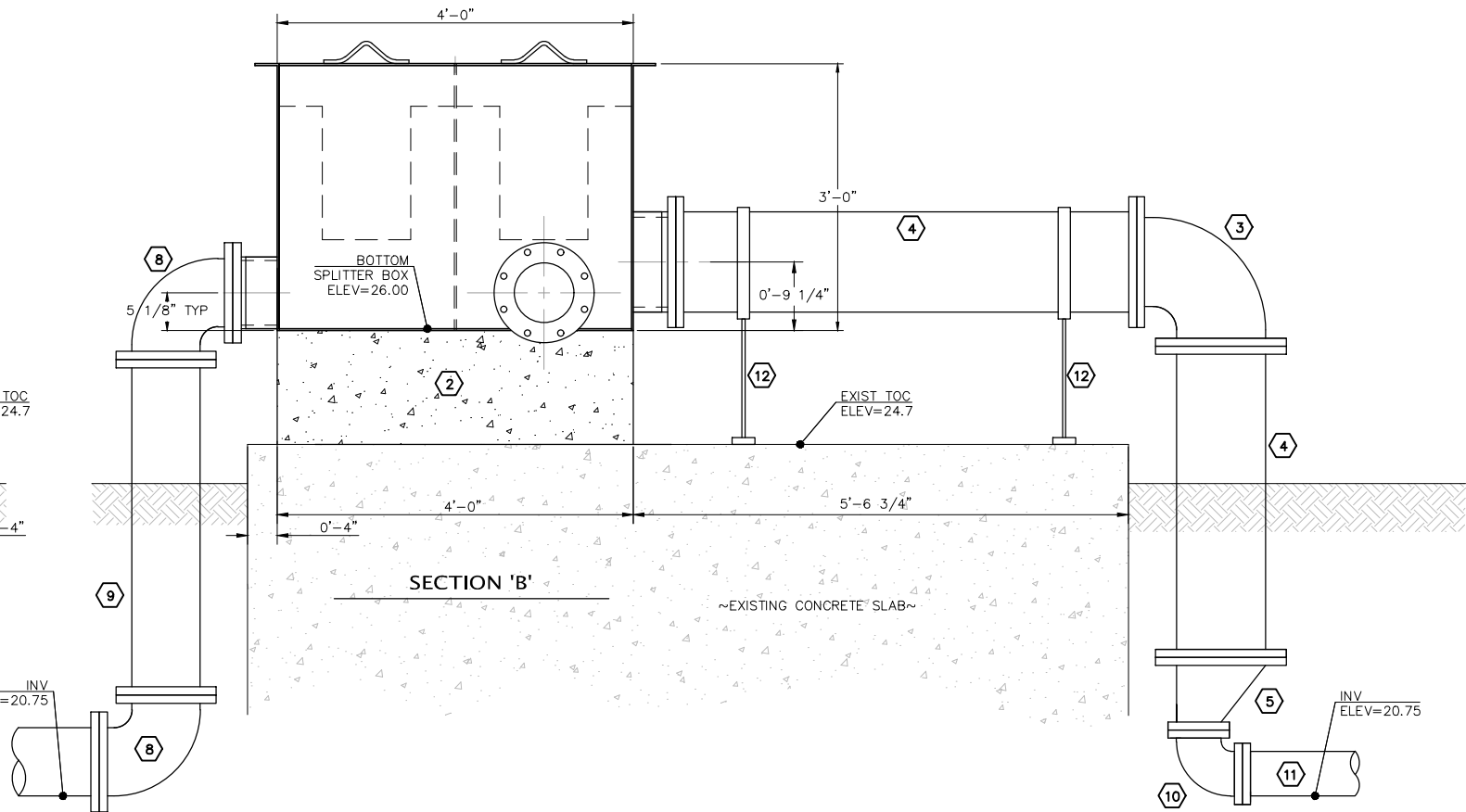
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- KEYED NOTES**
- (1) = FUTURE CONNECTION. SEAL WITH BLIND FLANGE
- (2) = PROPOSED CONCRETE SUPPORT FOR SPLITTER BOX. APPROXIMATE DIMENSION = 6.5' X 4.0' X 1.3' TALL. CONTRACTOR TO COORDINATE WITH ACTUAL SPLITTER BOX DIMENSIONS, AND ENSURE TOP OF CONCRETE ELEV=26.00. SEE STRUCTURAL DRAWINGS.
- (3) = 12" DI 90° BEND, FL
- (4) = 12" DI FLANGE SPOOL
- (5) = 12"x6" DI ECCENTRIC REDUCER, FL
- (6) = 6" DI 90° BEND, FL
- (7) = FLOW SPLITTER STRUCTURE IS TO BE A PREFABRICATED UNIT CONSTRUCTED OF 316 SS AND SIZED TO MATCH DRAWING UNIT IS TO INCLUDE 4 EQUAL SPLITTER CHAMBERS FOR FLOW TO EACH OF THE 3 TRAINS AND 1 ADDITIONAL TRAIN. UNIT IS TO INCLUDE A REMOVABLE COVER. EACH OF THE 4 FLOW SPLITTER CHAMBERS IS TO INCLUDE AN ADJUSTABLE WEIR. MFR TO BE SMITH & LOVELESS OR APPROVED EQUAL.
- (8) = 8" DI 90° BEND, FL
- (9) = 8" DI FLANGE SPOOL
- (10) = 6" DI 90° BEND, RESTRAINED
- (11) = 6" DI PIPE, RESTRAINED
- (12) = 12" PIPE SUPPORT PER $\frac{5}{C7.2}$
- (13) = 8" PIPE SUPPORT PER $\frac{5}{C7.2}$